



PUZeN e Booster System Installation and Operating Instructions

MODEL

PZEU Series

PZEC Series

PZEM Series

PZEH Series



Please read your instruction manual carefully as it provides instructions on safe installation, use, and maintenance. The appearance and specifications may be changed without notice to raise the unit's quality.

Contents

Thank you for purchasing WILO Pumps products.

You can use more conveniently after reading
these operating instructions

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Introduction

- ▶ These instructions for use contain information on PUZeN e booster system. Before using your booster system, you should read and understand the entire instructions. Make special note of all danger, warning and caution statements. Keep the instructions for using as a reference.
- ▶ If not observing these operating instructions, a serious injury or a death could happen by electric hazard. Also, install spare products at the special place where serious damages could be happened by no water supply caused by product troubles.
- ▶ Move together these operating instructions and a product when changing the location of the product. In this case, the using purpose and installing condition should be appropriate. Use after necessary treatments (example, Reset the operating data)
- ▶ Don't use except the normal purpose and installing conditions. If not, It could cause troubles or accidents. In that case, customers could have responsibility about that.
- ▶ If you have any question or error of these operating instructions, please contact to our agents or the head office.

Characteristics of product

- ▶ Applying to new technology in booster system.
- ▶ Optimum structure through the Controller module.
- ▶ Minimize pressure fluctuations through multi FC
- ▶ Offering RS-485 Modbus communication port.
- ▶ Automatically distribute load to each pump.
- ▶ 7" full color touch screen as standard.
- ▶ Can control max. 6pumps

Application

- ▶ Pressurized water supply and pressure boosting in residential, commercial and public buildings, hotels, hospitals, department stores and for industrial and process engineering.
- ※ Only use this product with clean water!
- ※ Excessively high or heavily fluctuating inlet pressure will require the provision and installation of a pressure reducing valve to maintain a constant inlet pressure level.

Specification

Refer to the name plate about capacity, pressure head, voltage, power of a booster system. Standard specification are as follows. (Some specifications could be changed by option)

- ▶ Product classification PUZeNe Series : Inverter control system

Liquid	Clean water(0~70 ℃)
Surrounding temp.	0~40 ℃
Humidity	Under RH 85% (Over dew point)
Pump type	Centrifugal pump

Location	Indoor
Suction condition	Pressurized
Protector for a motor	Over load relay
※ Pump power source	3Phase, 380V, 50Hz (Standard)

※ Power source could be changed as each product. Therefore, must check the name plate surely. 3

Safety precautions (Make sure to read & follow carefully.)

These instructions contain basic references which must be strictly adhered to. It is imperative for the installer and the operator to carefully read these instructions prior to installation and commissioning. Please observe the safety directions under the main heading and following those.

1 Safety marks contained in these instructions.

The following safety marks explain the grades of human and property damage when users ignore the instructions.

 Warning	When not following indications marked by the warning signal Death or serious Injury could happen.
 Caution	When not following indications marked by the caution signal Human damage or property damage could happen.
	This signal warns the electric hazard. Therefore, follow the indications of the warning signal or the instructions carefully.

2. Personnel qualification

The personnel installing the pump must have the appropriate qualification for this work.

3. Dangers if the safety indication are not followed.

Failure to observe the safety indications can endanger persons and the installation. Failure to abide by the safety indications will nullify your right to any claims for compensation. In particular, failure to observe these indications can create the following dangers:

- Failure of important functions in the installation.
- Dangers to persons caused by electrical and mechanical effects.
- Environmental pollution by dangerous handling solution leakage.

4. Safety instructions for the operator.

Local regulations for the prevention of accidents must be observed. Dangers resulting from electrical energy are to be prevented national/local energy supply company regulations must be observed.

5. Safety indications for inspection and assembly work

The operator must ensure that all inspection and assembly is carried out by authorized and qualified specialist staff who have studied the operating instructions in detail. In principle, work may only be carried out on the installation when it is at rest.

6. Unauthorized modification and manufacture on spare parts

Changes to the pump/installation may only be made in agreement with the manufacturer. The use of original spare parts and accessories authorized by the manufacturer will ensure safety. The use of any other parts may invalidate claims invoking the liability of the manufacturer for any consequence.

7. Improper operating modes

The operational safety of the appliance is only guaranteed if it is used as intended in accordance with the operating instructions. The operational tolerance limitations stated in the catalogue/data sheet must not be exceeded under any circumstance.

Transport and intermediate storage

Make sure if there are the damages of products on transport or not. If you find any damage, contact us. Deliver productions which have been properly packed.
Don't give any impact on products

Warning  Transport of the unit requires proper preparation and handling. Always make sure that the pumps and unit remain in horizontal position during transport and cannot slip out of the transport suspension arrangement. If the pump/unit slip out of the suspension arrangement, It may cause personnel injury and damage to property!

Caution  Avoid a extreme variation of temperature and humidity when keeping at a warehouse. Condensed water caused by severe humidity could cause the damage of metal components
Admitted temperature = -10 ~ 50 °C / Admitted humidity = under RH 85 %

 Be careful not to stress on cables of motor on delivery!
If cables are damaged, it could cause a electric shock, a leakage and a fire.

Components of product

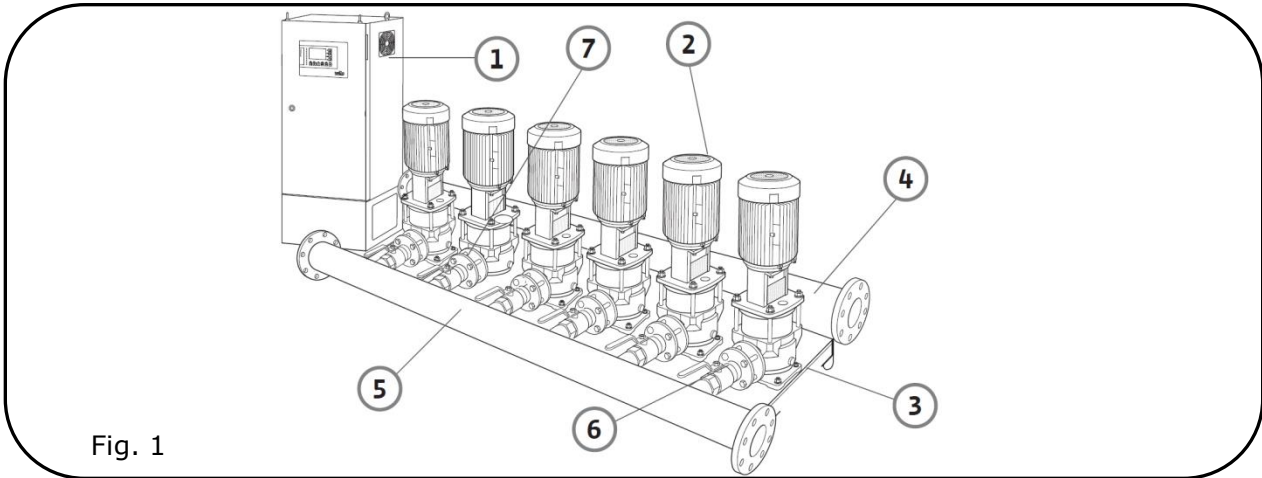


Fig. 1

No.	Components	Amounts
1	Control unit	1
2	Pump	2~6
3	Bed	1
4	Manifold for suction	1
5	Manifold for discharge	1
6	Isolating valve	2/pump
7	Check valve	1/pump

•Shape of components could be changed according to models.

Installation

Check whether products are complete or not!

- Ascertain the same product as you've ordered by a name plate and perhaps, if any, damage on delivery.
- Ascertain all components of pumps are complete. (refer to "components of products")



A booster system is used with a pressure tank which has proper capacity and is connected to the discharge manifold. Especially, the step control type must use a pressure tank, unless, it could cause failures by severe pressure fluctuation and frequently repetition of the start and stop.

1. Install location

- A booster system is for the indoor, nevertheless, if installed outdoor, make devices to avoid rain, wind and damage by freezing.
- The suction of a pump should be pressurized. (recommended suction pressure : 0.2kgf/cm²)
- Don't put the hands or foreign matter such as stick. Serious personal injury and property damage may result.



Water level of a reservoir always keep higher than the top of a pump's suction for avoiding dry operation. Keep over recommended suction pressure at the suction of a pump. Especially, must vent air after cleaning a reservoir entirely! And fluctuation of the suction pressure should be below 1.0kgf/cm²

- Install where to be without direct sun light, dirty & humidity and corrosive and inflammable gas whereas, to have good ventilation. (Admitted surrounding temperature is 0 ~ 40℃.)
- Install where easily to repair and check. Make booster system handled by a qualified personnel.
- Install where is close to water source and the length of a suction pipe should be short.
- If there is a possibility to be frozen to burst, Use complementary goods for avoiding freezing.



Obviously, make facilities for drain to avoid submergence damage by water leakage on installation, replacement, repair, and pump's failure!

2. Plumbing

- Make suction/discharge pipes not weigh a booster system
- Make a suction pipe short and not bent as much as possible.
- Attach the strainer for prevent pumps from a foreign matter interrupting normal pump operation.
- Recommend to attach flexible joints or valves at a suction/discharge pipe conveniently to disassemble.
- Recommend to install by-pass pipe at a discharge pipe for conveniently checking.
- Equip device to avoid water hammer at discharge pipe.
- If you operate it when damaged pressure tank, it is possible that discharge pressure is irregular or has leak.
- In near the confluent pipe, don't bend the discharge and suction pipe. It may be cause of noise and vibration.



Enlarge a suction pipe over one level more than standard in case suction is over 10m or bent over two times. If not so, the fluctuation of discharge pressure could be severe or the protector of dry operation could work at suction side. Also, do as the above in case that electric valves attach to discharge or water demand severely!

3. Electric wiring

Electric Wiring should be performed by electrician having a license. In order to prevent electric shock, the breaker against electric leakage and the grounding should be installed.

- Electric wiring about the main parts should be completed on installation site for avoiding damage during the transport.
- The main (R.S.T) and earth (E) should be wired in accordance with the wiring diagram inside of the control unit.
- When moving and installing the panel, bring each sensor with it to prevent extension electrical wire. And don't wire together power and sensor wiring(More than 10cm)
- Don't connect the earth wire with gas pipes, or it could be the cause of an explosion.

Installation

- Please check the following items

- ① Is a breaker for the electric leakage proper?
 - ② Is the electric wiring proper? (about the connecting state and the diameter of lines)
 - ③ Notice that three terminals of a motor should be firmly assembled.
- Please check the rotating direction of a pump! The direction is marked on a motor fan cover or a pump. Don't run the pump reversely.
 - Don't allow the power wires to come into contact with piping, pumps and motors.
 - The power source should be in accordance with the name plate.

4. Dry running protector

Booster system may occur fatal damage in driving without water.

Dry running protector is equipped to avoid pump's dry running.

On the improper suction condition, a electrode bar could work abnormally.

Before installing, please check whether the suction condition is proper or not!

1) Protecting dry running with electrode bar

Through electrode bar, you can use it without any additional setting.

At the Suction, a electrode bar is assembled for avoiding dry operation

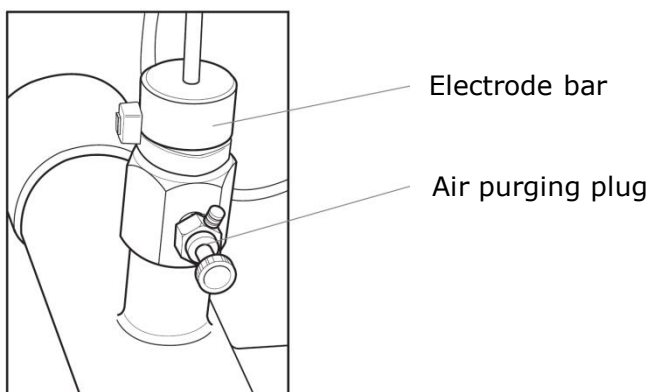


Fig. 2 Dry running protector

(The place impossible to use a electrode bar)

If a suction pipe is too long or bent too many times, the function against dry operation could work, because air is suctioned together with water on operating.



Caution

If abnormal noise happen at the suction, stop the system, and after improving the suction condition, operate a system again.

Don't use the system at the place where air is apt to be suctioned with water.

If not, the components for rotating are apt to be broken.

Installation

2) Preventing low suction pressure by pressure transmitter

Preventing dry running by receiving the signal from the transmitter and sensing the suction pipe pressure with pressure transmitter method.

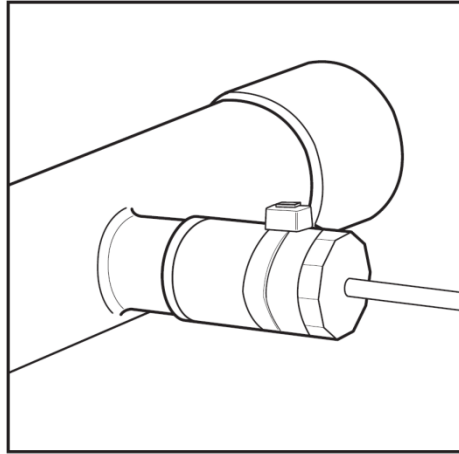


Fig 3. Pressure transmitter in the discharge pipe

5. Pressure transmitter on discharge side

pressure transmitter on discharge side to drive booster system.

Two of sensors are installed on discharge pipe for main sensor and spare sensor function.

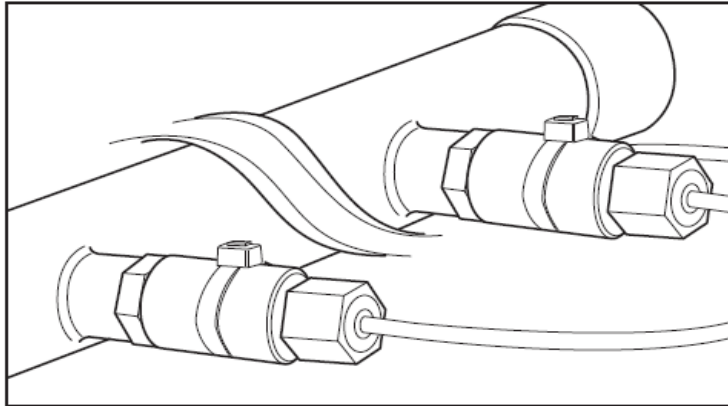


Fig 4. Main and spare sensor by using two pressure transmitter

Installation



Don't ramify pump's terminal for power supply to use another purpose for it. That is the reason for system fault.
Don't come metallic foreign substance into internal control panel during the electric wiring.
Don't exceed $\pm 10\%$ of working voltage.

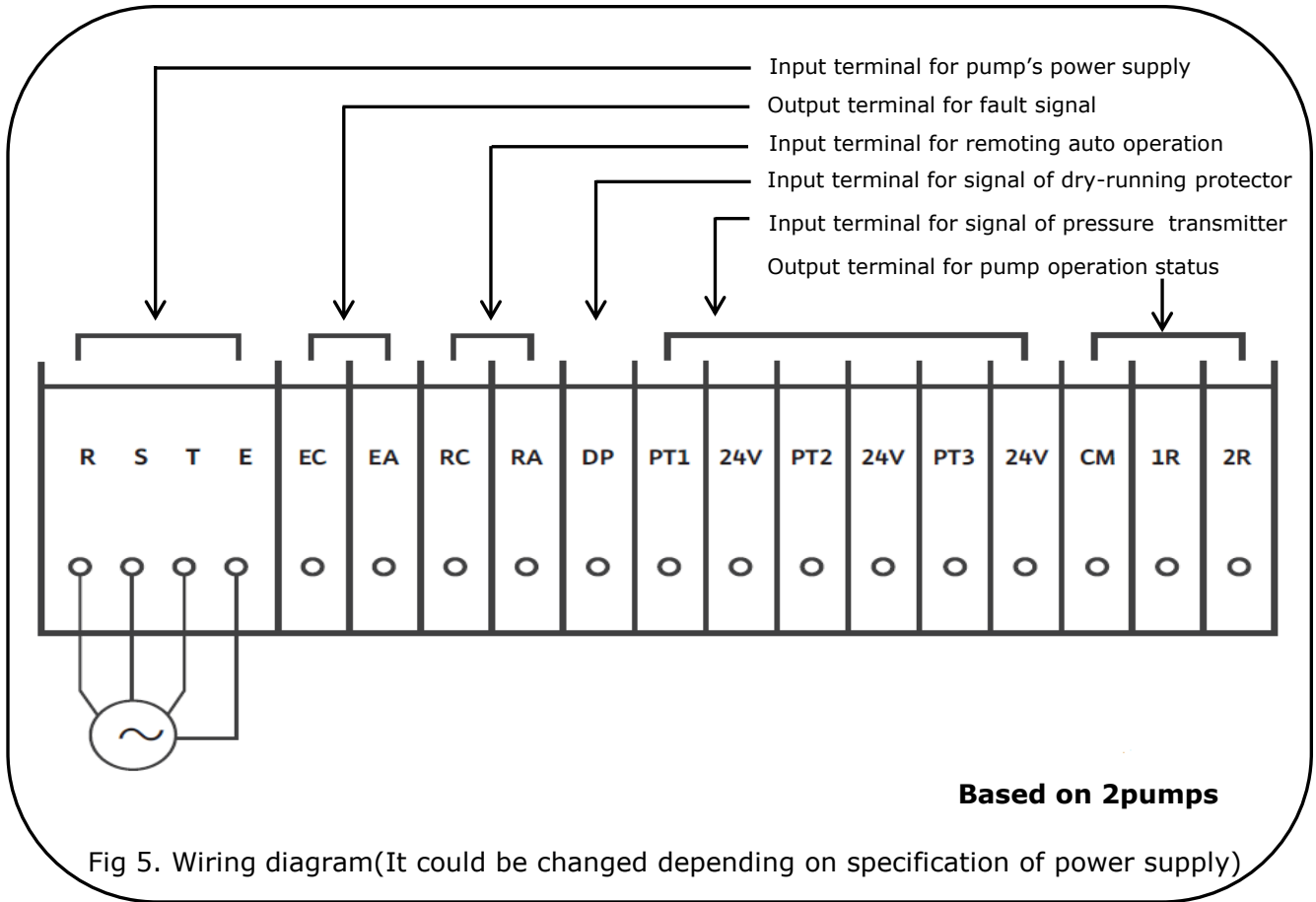


Fig 5. Wiring diagram(It could be changed depending on specification of power supply)

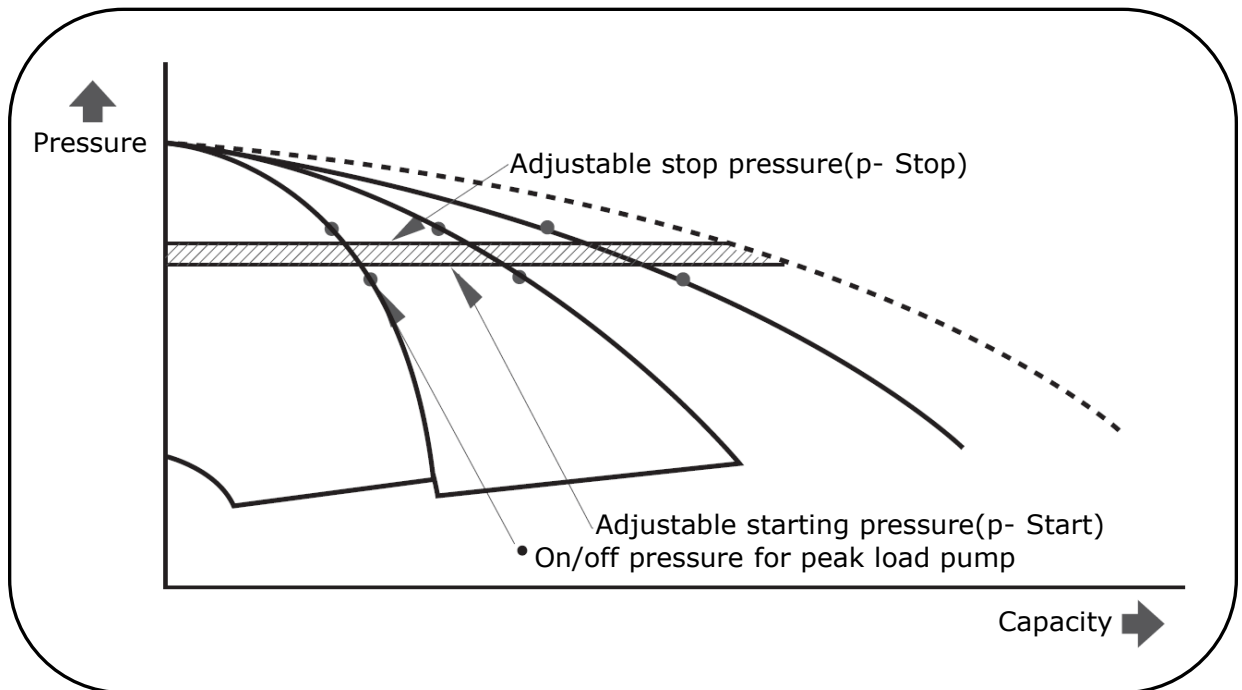
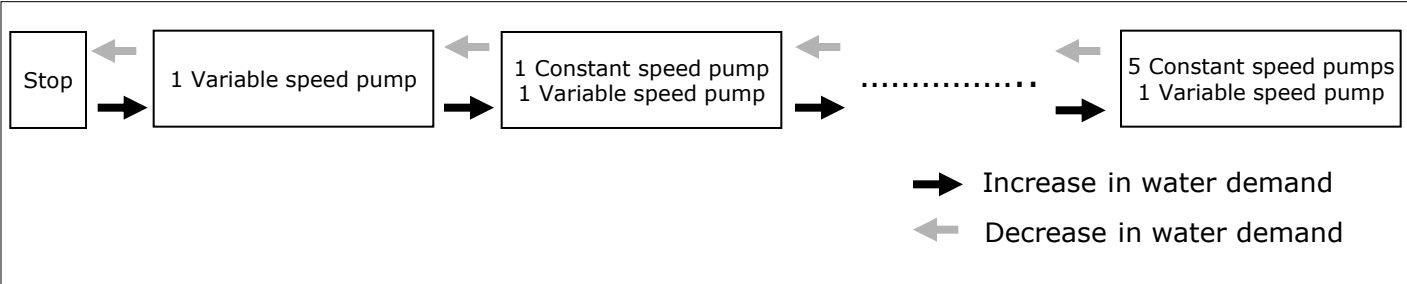
Name of terminal	Function
R, S, T, E	Input terminal for pump's power supply(Standard: 3Φ 380V 50Hz)
EC, EA	Output terminal for fault signal In case that booster system occurs some fault, output fault signal as electric contact. If the fault is cleared, automatic reset. Normal condition: EC-EA(OFF) Fault condition: EC-EA(ON)
RC, RA	Input terminal for signal of remoting auto driving
DP	Input terminal for water level(electrode bar) signal of dry-running protector
24V, PT1, PT2, PT3	Input terminal for signal of pressure transmitter
CM, 1R, 2R	Output terminal for pump operation status

* Composition for terminal is changed depending on specification of power supply, option for control panel.
* Refer to wiring diagram attached on each control panel.

Operating principle

1. Single Inverter operating mode, PZEU Series

- Operating the pumps stand in a row from 2 to 6.
- Depending on the sequential operation, all pump(among the selected by pump setting status with no abnormalities) evenly operate.(To prevent over-load of the one pump)



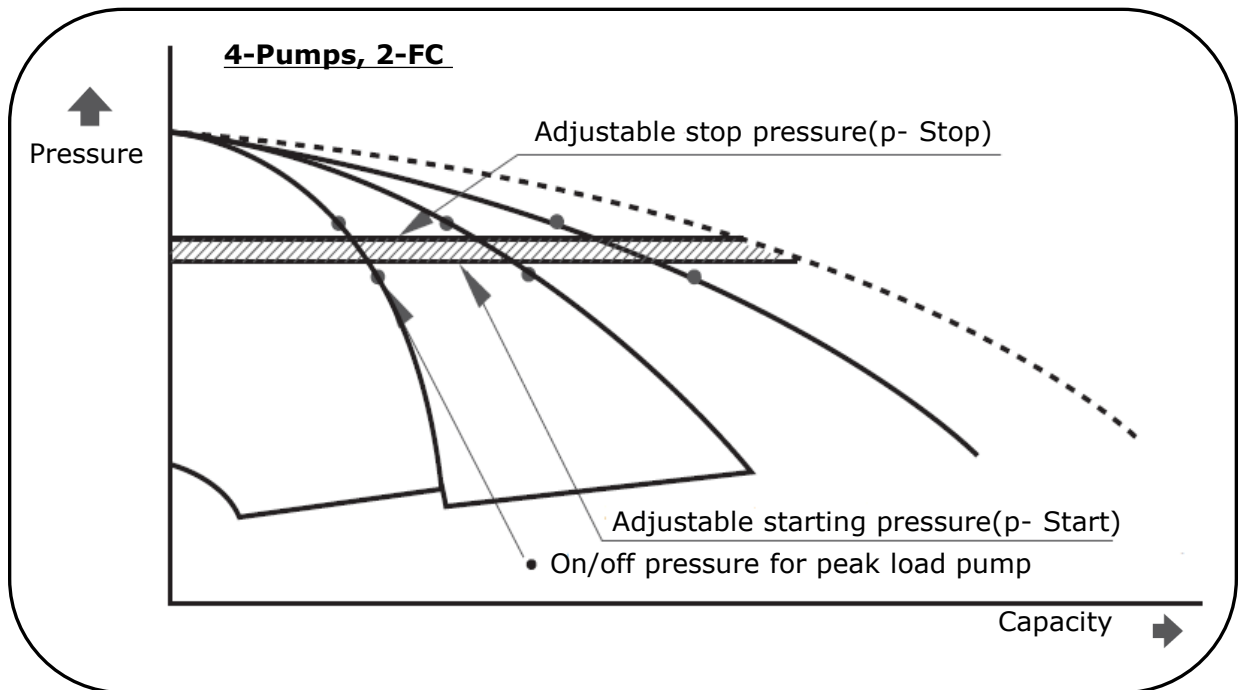
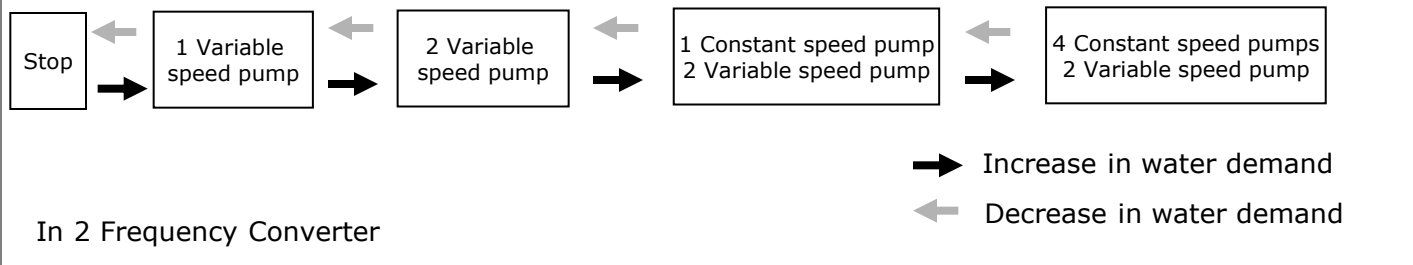
<Operation condition>

- ① When current pressure goes down below set-pressure in discharge parts, FC pump is operating initially.
- ② If the capacity is continuously increasing, FC pump will be operating to max. speed. When the FC pump reach max. speed, constant speed pump is operating in a row.
- ③ If the capacity is decreasing, FC pump will go down to speed by starting frequency. Constant speed operating pumps will be stopped and FC pump keep set-pressure constantly.
- ④ If there is no capacity, FC pump will raise current-pressure by stop-pressure and then stop operating.

Operating principle

2. Combi Inverter operating mode, PZEC Series

- Operating the pumps stand in a row from 2 to 6.
- FC pumps are set according to attached FC numbers.
- The operation of the FC pumps is given priority depending on setting.
- Constant speed driving pumps are operated by sequential.(To prevent over-load of the one pump)



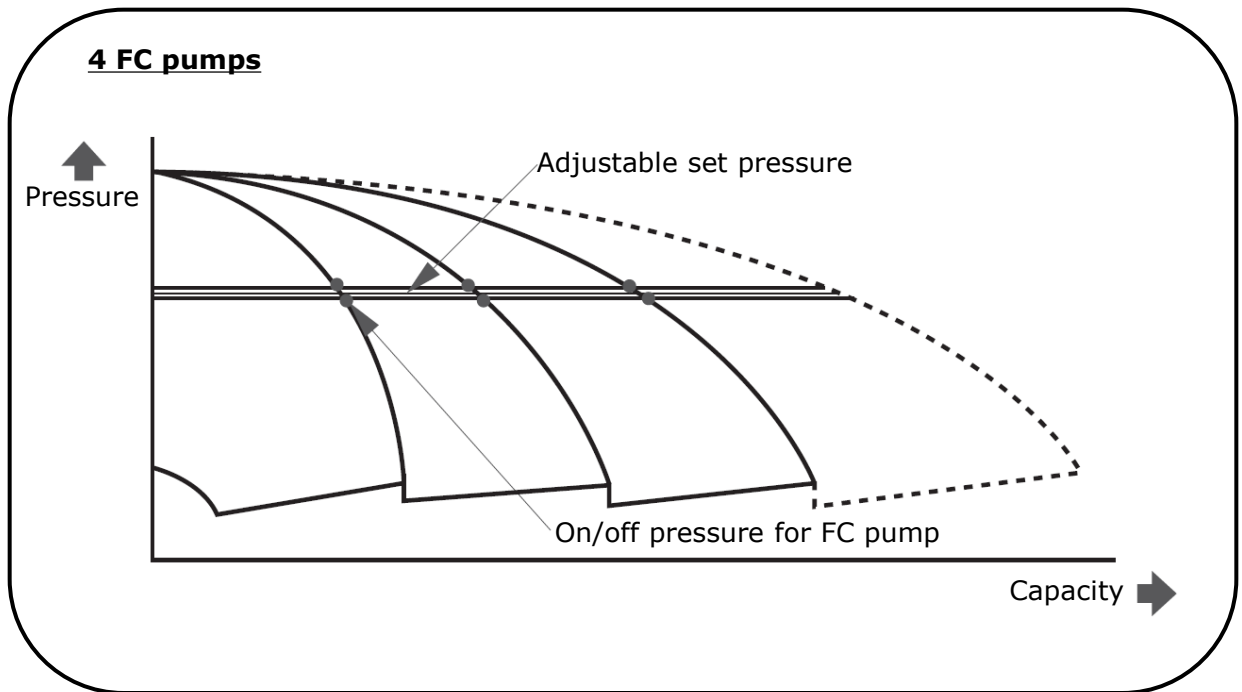
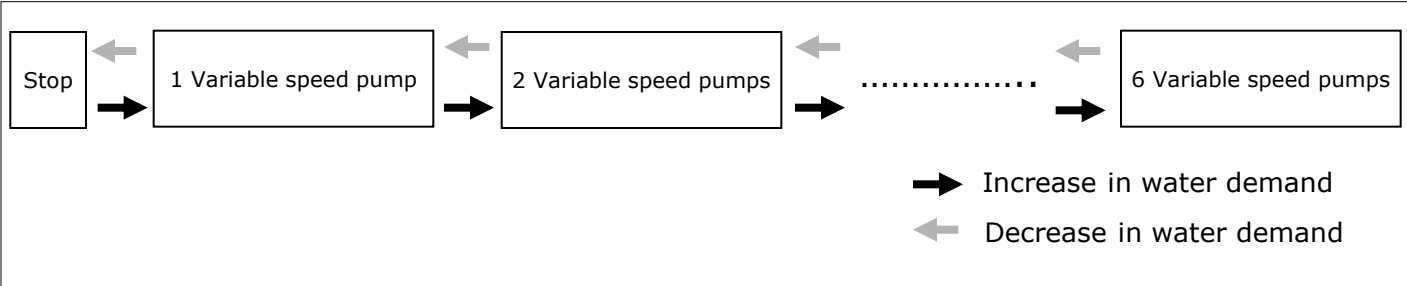
<Operation condition>

- ① When current pressure goes down below set-pressure in discharge parts, FC pump is operating initially.
- ② If the capacity is continuously increasing, FC pump will be operating to max. speed, and second FC pump will drive. It becomes main pump.
- ③ If the capacity is increasing, constant operation pumps drive in turn.
- ④ If the capacity is decreasing, FC pump will go down to speed by starting frequency. Constant speed operating pumps will be stopped and FC pump keep set-pressure constantly.
- ⑤ If the capacity is decreasing continuously, main FC pump stops driving and the next FC pump is set as the main pump and then operate.
- ⑥ If there is no capacity, FC pump will raise current-pressure by stop-pressure and then stop operating.

Operating principle

3. Multi Inverter operating mode, PZEM Series

- Operating the pumps stand in a row from 2 to 6.
- All of the pumps are operating through FC. Operation is set by step control.



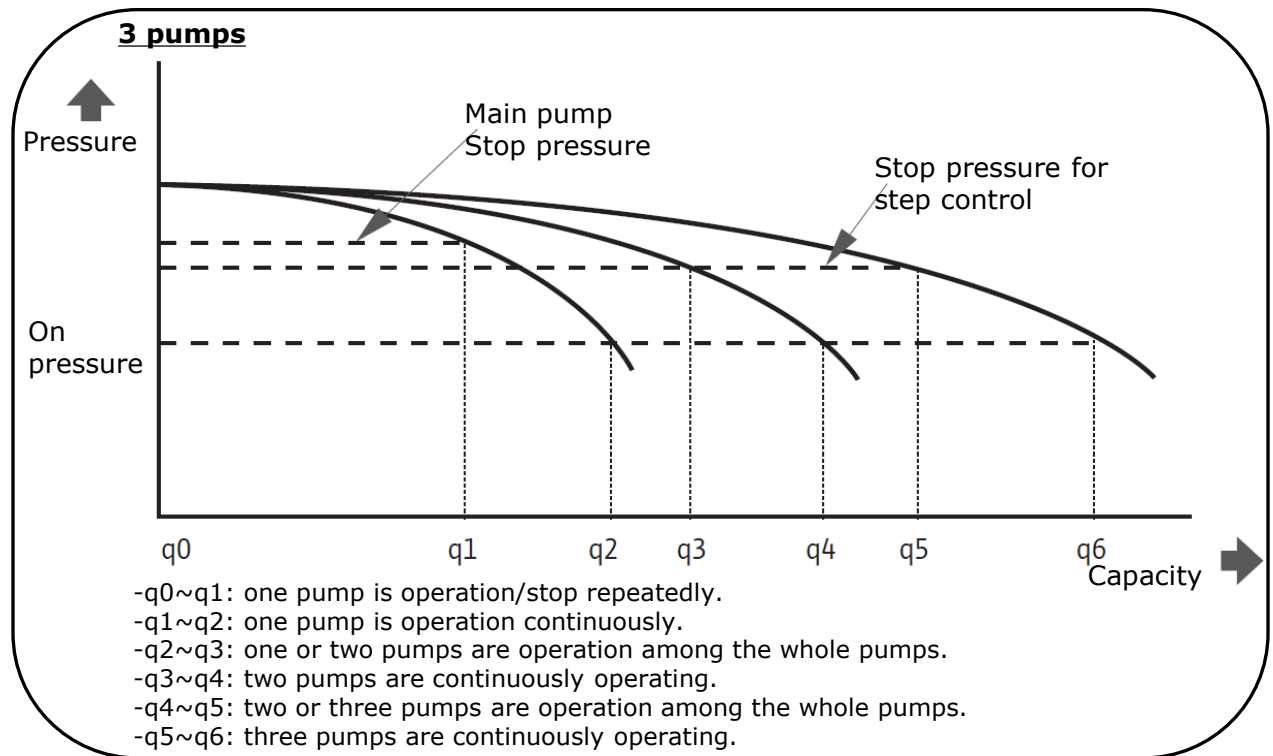
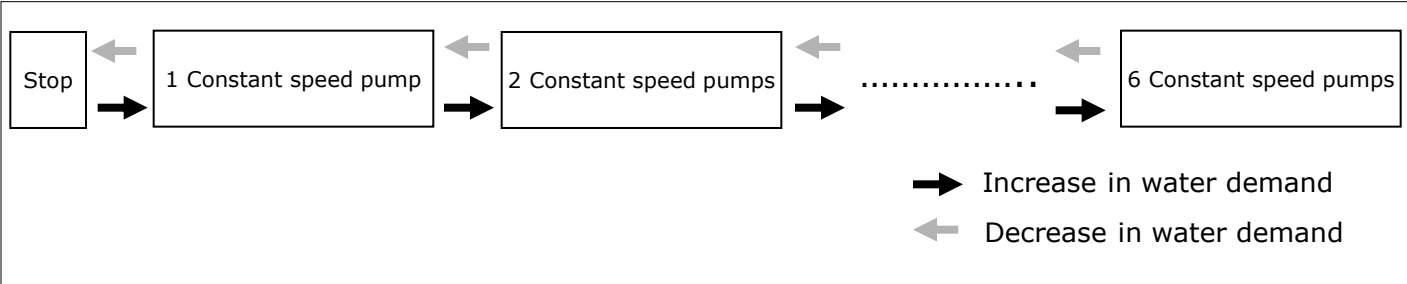
<Operation condition>

- ① When current pressure goes down below set-pressure in discharge parts, FC pump is operating initially.
- ② If the capacity is continuously increasing, Initial FC pump will be operating to max. speed. When the FC pump reach max. speed, second FC pump will operate. It will be main FC pump.
- ③ If the capacity is continuously increasing, Second FC pump will be operating to max. speed. When the FC pump reach max. speed, third FC pump will operate. It will be main FC pump.
- ④ If the capacity is decreasing, third FC pump is stopped and second FC pump will shift to main pump.
- ⑤ If the capacity is decreasing, the main FC pump is stopped. The other FC pumps will shift to main pump. Main FC pump is maintaining the set pressure.
- ⑥ If there is no capacity, FC pump will raise current-pressure by stop-pressure and then stop operating.

Operating principle

4. Step operating mode

- Operating the pumps stand in a row from 2 to 6.
- All of the pumps are operating by step control if there are no problems in the pumps.
(To prevent over-load of the one pump)



<Operation condition>

- ① When current pressure goes down below set-pressure in discharge parts, one pump is operating initially.
- ② If the capacity is continuously increasing, the other pumps are operating by turn.
- ③ If the capacity is decreasing, initial operating pump will be stopped.
- ④ Pump will be stopped, when it reach following conditions.
 - ▶ If the pump is operating, pump will not stop over the stop pressure; It should be past "Min. run time".
 - ▶ Currently several pumps are operating. If you want to stop it, "Seq. delay time" should be past.
 - ▶ Merely, pump's auto operation switch turns off or has a trouble which is stopped. In this case, "Min. run time" does not use, it will follow "Seq. delay time".
- ⑤ If there is no capacity, last operating pump will raise set pressure and stop.

Operation



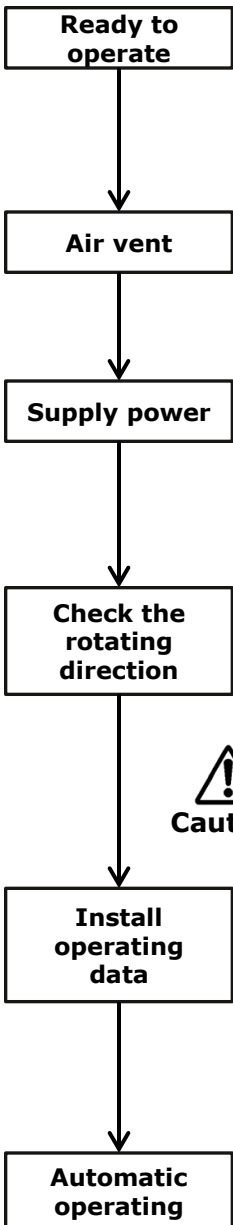
Close the door of a control unit on operating for avoiding electric accidents by high voltage and hazards caused by the leakage of pumps and pipes. Besides, electric appliances around a system should be treated in the same way.



Care not to block the motor ventilator with unexpected materials before operating. It could cause a fire, damages of a product.

Please Re-drive after turning off all lamps when stopping booster system.

1. Initial operating order



- ① Check a main power breaker and a protector capacity.
- ② Check the cable square of main power.
- ③ Check a main power voltage.
- ④ Check connecting state of pipes and wires.
- ⑤ Check water level in a suction reservoir.

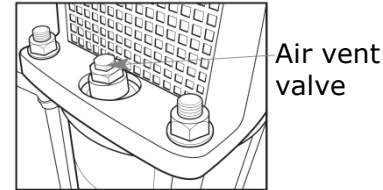


Fig 6. Air vent valve

At each pump, close a isolating valve in the suction and release a air vent valve(Fig 6.) and slowly open a isolating valve in the suction. Water and air come out together. If air vents sufficiently, fasten the plug again and open the isolating valve completely.(The operation with a isolating valve close could cause dry operation and troubles.)

- ① Supply power (Pump power / Control power)
 - ② Turn on all breakers (MCCB & ELCB) in a control unit.
- At that time, pump operating switch(page19,6) must be placed "Stop" .

Check the rotating direction of a pump as operating each pump in a short term by pump operating switch(page19,6) placed "Manual". the rotating direction of a pump is marked on the motor fan cover or the pump. If the direction is reverse, change two of pump power lines (R.S.T).



If you separate panel and motor's wiring, CONNET it in same order as before separation. Otherwise, it will be cause of fault about different direction of rotation FC and direct operation.

Set operating data for the automatic operation.
(The setup of operating data for each items refers to 『Set』

✎ Essential operating data :

Setting Pressure, Final Stop Pressure, Inverter Minimum frequency, Shut-Off Pressure

If pump operating switch(page19,6) is placed "Auto" and touching "Auto" icon on the 7" touch screen display, the system is operated in accordance with operating data. Please, check operating state and whether it work normally or not.

At initial operation, Must vent the air which stayed inside pipes or pumps as opening an air vent plug.
If air does not vent sufficiently, even if opening a tap, water could not come out.
On operating, check whether pressure, vibration, and noise are normal or not.

Maintenance



Warning

When repairing or checking the system, must cut off all powers and make pumps stop for the safety. On automatically operating, there is the hazard that the system could be operated unexpectedly.

When flooded the product, there is concern about electric shock. You must drive after repairing and safety checking through WILO service center.

1. General check

- In case that on-pressure, off-pressure, electric current and noise are higher than normal value, refer to 『trouble - shooting』.
- If a mechanical seal is normal, there is almost not leakage.
If the leakage is too much, replace it with new one.
- Check up the leakage and the damage of pipes.

2. Adjustment of the on-pressure

Adjusting of the on-pressure only in case that the discharge pressure is too high or low

3. Maintenance in the winter

In the winter, when not using for long time to avoid being frozen to burst, use thermal insulations for pipes and manifolds or drain water from pumps and manifolds through plugs.

4. Charging the pressure tank.

On checking up the pressure tank, if the pre-loading pressure of a tank is too low, charge air as the following way. After completely draining the water in the tank, open the cap on the top of a tank and charge air with a air compressor. Be careful not to charge the oil, which is mixed with air, to the pressure tank. If the pre-loading pressure is too low, the pressure tank does not normally work. Oppositely, if the pre-loading pressure is higher than the on-pressure, the on/off of the pump is too much repeated. in that case, vent air as pressing a part popped out in the center of a air valve.

**※ The pre-loading pressure is set as the 90% of on-pressure or
" on-pressure - 0.5kgf/cm² " (deviation ±10 %)**

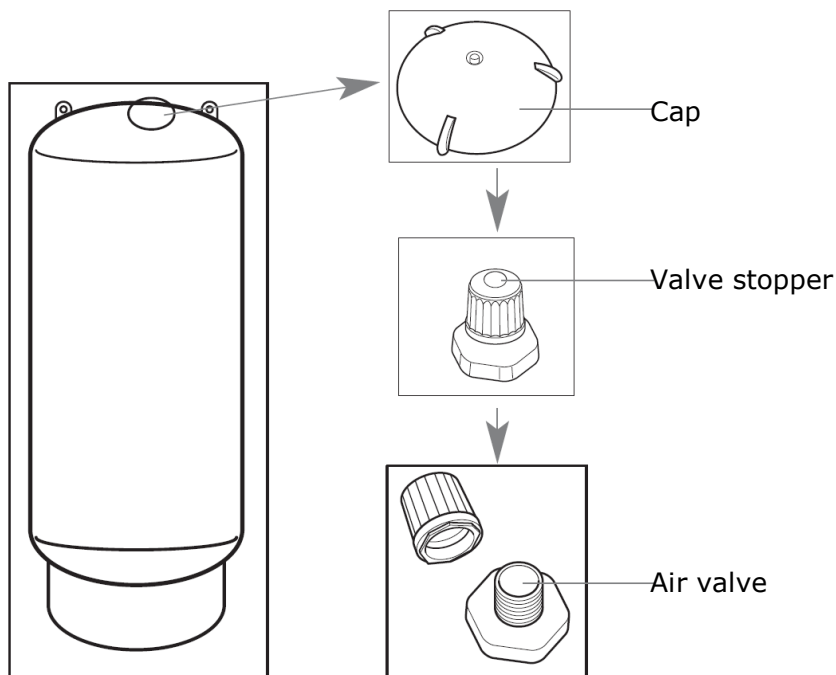


Fig 7. Membrane tank

Maintenance



5. Checking the control unit

**Don't arbitrarily change the structure and installation except the operating data.
The personnel checking the control unit must have the appropriate qualifications.
Must turn off the main power before the inspection or the maintenance of a control unit.**

- Carefully check up the following items periodically.

- ① The cooling state of a inverter.
- ② The overheat, the abnormal smell, the color deformation of a part in a control box.
- ③ The state of wiring.
- ④ The vibration and the noise.

- Never do a insulation test nor a withstand voltage test. It could breakdown the semiconductors in the control unit.

- The static electricity that is occurred by touching, damages the electric circuit. Therefore, must handle the circuit board with care.

- In a inverter type, even after the main power has already been off, it takes 3~4 minutes to completely discharge a inverter. Therefore, wait until the inverter is completely discharged.

- After turning off the power, check the rust of bolts and nuts. Tighten loosened screws and nuts.

- In case of a inverter type, when there is a foreign matter by the cooling fan of a inverter, cooling capacity becomes down. Therefore, removes it, after turning off the main power and completely discharging the inverter.

- Check up the rotating troubles of cooling fans, the troubles of magnetic contactors, the abnormal noise, and if there is any trouble, replace it with new one.

6. The replacement time of major components

The expected life of each component is calculated on the normal operating condition. Replace the component by the following criteria.

Classification	Component	Life time	Criterion of replacement
Pump & motor	Mechanical seal	1year	Excessive leakage
	O-ring and gasket	Each time	Whenever disassembling
	Motor bearing	3years	Abnormal noise or severs noise
Mechanical parts	Pressure transmitter	3years	When output signal is not correct
	Pressure tank	3years	Abnormal operation(Pressure fluctuation)
Control unit	Magnetic contactor	3years	Severe contact's damage or abnormal operation
	Relay	3years	Severe contact's damage or abnormal operation
	Inverter electrolytic condenser	3years	When capacity is reduced
	Inverter cooling fan	2years	Abnormal noise
	Circuit board	3years	Uncertain performance

- **When disposing a printed circuit board or electric components, observe the relevant law!**
- **Our product include a printed circuit board. Don't dispose it at a common garbage can!**

Trouble-shooting

1. General faults

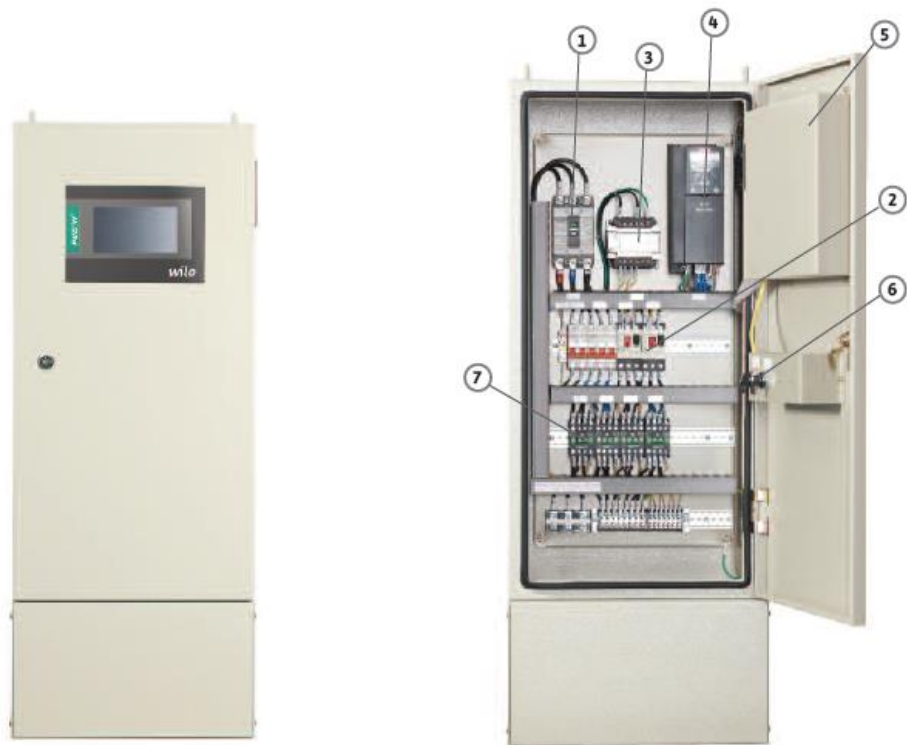
Fault	Cause	Remedy
Even if a tap is open, water does not flow.	• A discharge valve is closed. • It is not enough to vent air. • Trouble in a pump/ a motor. • A motor rotates reversely. • Water level of a reservoir is low. • Trouble in a pressure transmitter. • Operation mode is in the "STOP". • Trouble in wiring(Disconnection) • On-pressure is low.	• Open a valve. • Vent air through a plug. • Repair or replace. • Exchange two of three main power terminals in a control unit. • Compensate water to a reservoir. • Repair or replace. • Change a operation mode to the "AUTO". • Repair or replace. • Reset the on-pressure.
Even if all tape are closed, pump don't stop.	• Off-pressure is high. • Trouble in a pressure transmitter. • The operation mode is "Manual" • Trouble in a control unit. • Leakage in pipes. • The performance of pump is low.	• Reset the off-pressure. • Replace. • Change the operation mode to "AUTO". • Repair or replace. • Repair or replace. • Repair or replace.
Pump is running without water flow.	• Leakage in pipes. • Leakage in a check valve.	• Repair or replace. • Repair or replace.
Vibration of pump is abnormal.	• The gap of on & off-pressure is small. • Trouble in a pressure tank. • Pressure fluctuation in pipes.	• Reset a on & off-pressure. • Recharge air to the required pressure. • Repair pipes(remove air).
Water doesn't reach to the highest tap.	• On-pressure is low. • The selection of a system is fault. • Pump's performance is down.	• Reset a on & off-pressure. • Recheck water capacity & head. • Repair or replace.
Pump don't work by turns or in parallel.	• Trouble in a control unit. • Trouble in a pressure transmitter. • Trouble in a pump or a motor.	• Repair or replace. • Repair or replace. • Repair or replace.
The MCCB in a control unit is tripped.	• Short of wiring between pump and power. • A circuit wiring short or a damaged breaker • Short of power line.	• Repair or replace. • Repair or replace. • Repair or replace.
Pressure fluctuation is severe.	• Inflow air because a suction pipe is too long or the diameter is too small. • Operating without a pressure tank or the pre-loading pressure is incorrect. • Electric valve works or shifts on water demand is severely large.	• Reduce the pipe's length & enlarge the diameter of pipes. • Check pre-loading pressure of a tank and adjust it. • Install a pressure reducing valve and enlarge capacity of a pressure tank.

Operating Instructions of Control Panel



Construction of control panel

PZEU Type



No.	Part Name	Q'ty(pcs)	Purpose
1	Main Circuit Breaker	1	For cut-off main power
2	Molded Case Circuit Breaker	1~6	For protecting pump
3	Transformer	1	For supplying power to control
4	Frequency converter	1~6	For operating pump
5	PCU+G22	1	Controller
6	Pump operating switch	2~6	For changing pump's operating mode
7	Magnetic contactor	4~12	For operating pump

- Construction of control panel could be changed according to models.

Safety rules

1. General

Installation and commissioning by trained and qualified personal only!

1.1 About this document

These Installation and operating instructions are an integral part of the product. They must be kept readily available at the place where the product is installed. Strict adherence to these instructions is a precondition for the proper use and correct operation of the product. These Installation and operating instructions correspond to the relevant version of the product and the underlying safety standards valid at the time of going to print.

2. Safety

These instructions contain important information which must be followed when installing and operating. These operating instructions must therefore be read before assembly and commissioning by the installer and the responsible operator. Both the general safety instructions in the "safety precautions" section and those in subsequent sections indicated by danger symbols should be carefully observed.

2.1 Danger symbols used in these operating instructions

Safety precautions in these operating instructions which, if not followed, could cause personal injury are indicated by the symbol.



Warning

When warning of electrical danger with.

2.2 Qualification

Installer must have a qualification for the work.

2.3 Warning when not following safety instruction

If you don't follow safety instruction, people can be in danger and you can't get any warranty. You may meet following danger.

- Failure of critical function.
- A person's risk by electrical or mechanical impact

2.4 Safety instruction for product manager

- Obey the existing regulation for accident prevention.
- Remove the risk factor by electrical energy

2.5 Safety instruction for checking and installing work

Only technical expert that know well about instruction checks and installs the product. Please do basic work in stop status.

2.6 Unauthorized modification and spare parts manufacturing

- If you want to change the components, get permission manufacturer in advance.
- Using genuine spare components and authorized in manufacturer part is safe.
- If you use other components and what happens, you can't get warranty.

2.7 Not allowed operating method

- Only in accordance with the correct use instruction in the manual, work normally.

3. Transport and intermediate storage

The control panel is to be protected against moisture and mechanical damage caused jolts/impacts.

It should not be exposed to temperatures outside the range -10 C to + 50 C.

4. Scope of application

The PUZeNe is used to automatically control of pressure boosting systems.

It is used in water supplies of residential high rise buildings, hotels, hospitals, administrative and industrial buildings.

Product and Accessories

1. Functional description

PUZeN e system is controlled and managed by the PCU+G22 in conjunction with the respective pressure and transmitters. The unit pumps are started and stopped in cascade mode in response to demand variations by pressure-actuation within the range between the set pressure levels. Splitting the unit capacity onto a number of small pumps ensures constant adaption to the actual demand situation within the limits of the set pressure level.

2. Structure of Control System

- . PUZeN e system can control up to 6 pumps.
- . The design of control system configured by pump and frequency converter configuration as PZEU series(1 frequency converter), PZEM series(each frequency converter to pumps) and PZEC Series(2~3 frequency converter)

3. Operation Mode

3.1 FC operation

An electronic pressure sensor provides the actual system pressure value as a 4-20 mA current signal. The controller(PCU+G22) then keeps the system pressure constant by comparing the desired and actual values. The speed controlled base load pump will start up if required. Here the speed of the pump depends on consumption. If the power requirements of this pump can't be met, a second pump is switched on. With increasing requirement further pumps will be switched on. If the requirement drops such that the controlling pump is operating in its lower power range and is not needed to meet requirements, one pump after the other switches off. At last the base load pump switches off by the zero flow detection. With decreasing system pressure under the set-point, the system switches on again.

3.2 Non-FC Pump operation

Operating method is almost same with FC operation but non-FC pumps are only operating with the constant speed.

4. Operating and stop of pumps

When just one pump is operating, it will be stopped after increases set pressure. If pressure doesn't decrease after the increased set pressure is returning to the original set pressure, basic load pump will turn off after the specified time.

When FC failure in PZEU and PZEC series, it automatically switches on step control.
(It is option in PZEM series)

Function

Optimal operation control for peak-load pump

The PCU+G22 optimize the peak-load pump on/off control.

Decrease unnecessary pump running and increase pump life cycle and save the energy.

Optimal speed control for variable speed pump

Reflect the law of similarity in program, when peak-load pump is on/off, minimize the pressure hunting.

Pressure Trigger

Display the set point pressure and actual pressure by real time graph.

Function

Indication of pump load rate

Display each pump load rate as circle graph.

Real time indication of pump operating rate

Display real time indication of pump operating rate.

Help

Display each function's description for user.

Change of pressure unit and power unit

Supply various pressure and power unit for user.

List of pump failure history

Maximum 37 alarms occur history can save and the alarm list used for booster system maintenance and prevent trouble.

Schedule operation

Automatically change the pressure set value according to hourly, monthly, weekly, seasonal schedule mode which can be set by users.

Standby pumps

The standby pump, which is defined by user, does not run during the normal operation. If the normal operating pump is stopped due to fault, the standby pump is cut-in.

Pump changeover

The pump with the lowest running hours is initially started and lastly stopped for the running time optimization.

Sequence delay time or Delay time for cut-in and out

For reducing pressure overshoot and the number of start/stop time of pumps, user can set the delay time(s) for pump cut-in and cut-out.

Stop at the small outlet water capacity(Stop pressure of main pump)

Main pump stops in small water capacity or stopping water capacity which can be detected by pressure sensor in a periodic time.

Max. pressure setting

Protect pipe system and pump system with auto-stopping for the unexpected pressure rising in outlet. System is automatically started when the actual pressure is increased up to the pressure set value after stop.

Min. pressure setting

For the preventing continuous running of system due to the leakage in the outlet piping system or pump system, the pump system stops when detecting the minimum pressure which can be set by user.

Function

Warning at abnormal low pressure

For the detecting of the leakage in the outlet pipe or dry running when the actual pressure is lower than pressure set value.

Ramp-up and -down time of frequency converter

For stabilizing of the output pressure and protecting motor, user can set the ramp-up and ramp-down time of frequency converter.

Min. frequency setting

To meet the required pressure quickly, setting the FC starting frequency.

Preventing of the pump blocking

The pump starts automatically and runs for a short time to prevent the pump blocking after the prolonged shutdown or the damage due to the operation with air in water.

Auto and manual operation

User can choose the operation mode either automatic or manual operation.
(PZEU Series standard)

Kick-off

The pump kicks off in case of fault.

Auto re-start

User can choose auto re-start or manual re-start when the electrical power is re-supplied after power failure.

Dry running protection

For ensuring from the damage of pump and shaft seal due to dry running, all pumps are stopped when the detecting of dry running through sensing device. After removing causes, system can automatically re-start or manually start. These starting methods can be selected in the menu. After over 5 times auto re-starting, system does not automatically re-start.

Locking password

Password can be set for the preventing from the unauthorized changing of the important setting values and operation parameters of system except the basic setting values.

PID control

Change the number of operating pump or speed as frequency with PID control in case of there is a difference between feedback value and set value.

Wired external controller

Externally control the pressure set value, start and stop of the system via serial communication(RS-485)

Function

Compensation for friction loss

Change the pressure set value to compensate the friction loss according to user's definition.

Differential pressure control

Through additional sensor and function, it can circulate hot water by differential pressure control(Optional).

Number of starting time

User can define the minimum running hours or operating counter per hour.

Diminishing(Reduction) operation

Pump system does not operate as maximum capacity in case of the electrical power shortages or water shortages.

Starting up

When you are starting up the pump, you should have service from the WILO service center. Before the first putting into power, performing moderate connecting wire. In particular, checking the connection. Starting up pump, you will check the whole of procedure. And refer to installation and operating instructions.



Re-tighten all supply terminals before putting the unit into operation !

1. Factory setting

The controller is preset factory side. This factory presetting can be restored again by WILO-Service.

2. Overcurrent protection

OCR(Over Current Relay)in MCCB or Frequency converter protect motor from overcurrent damage.

Fault message

When occur system fault, the cause of the fault must be removed.

Fault text	Causes	System conditions	Remedy
Pump # Fault	Overcurrent of motor.	Pump stop	Check the pump
Dry running	Detecting dry running by dry running protection.	System stop	Check suction condition
Repeated dry	Dry running occurred more than 5 time in the 24 hours.	System stop	Check suction condition
FC # Fault	Fault of FC.	System stop or run peak pump	Check # FC
Low discharge Pressure	Low discharge pressure even if all pumps running.	Continuously operating but still alarm	Check discharge condition
Setting values	Wrong parameters in set values.	System stop	Check setting values
PT Fault	Signal error from transmitter. And value of sub-pressure transmitter is different with main transmitter value over 20%.	System stop	Check pressure transmitter
Max. pressure	Over pressure than Max. pressure value.	System stop	Check discharge condition
Repeated Max. Pressure	Repeated Max. Pressure error occurs more than 5 times within 5 minutes.	System stop	Check discharge condition
Min. pressure	Low pressure than Min. pressure value.	Alarm or Stop	Check discharge condition
S-PT Fault	Fault of sub transmitter.	Only alarm	Check S-PT
M-PT Fault	Fault of the main transmitter. (automatically shift sub transmitter)	Only alarm	Check M-PT
10% PTs Error	Value of sub. pressure transmitter is different with main transmitter value over 10%.	Alarm	Check S-PT & M-PT
Low suction pressure	Detecting dry running by transmitter.	System stop	Check suction condition
# starting No.	Over the number of pump start/stop than set valve.	Alarm	Check the operation condition

Abbreviation

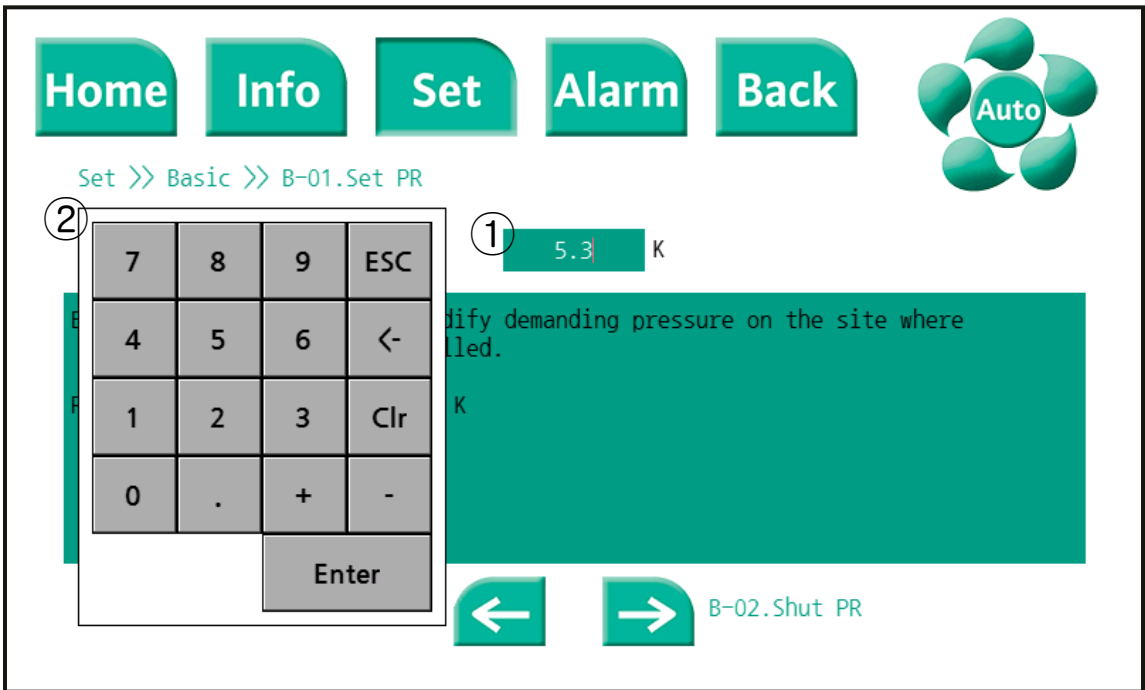
PR : Pressure
 PRE : Previous
 DIS : Discharge
 PT : Pressure transmitter
 SUC : Suction
 MIN : Minimum
 MAX : Maximum
 FC : Frequency converter
 FREQ : Frequency
 DTC : Detecting

Icon

Icon	Action
	Move to Home
	Move to Info
	Move to Set
	Move to Alarm
	Move to upper page
	Delete recovered alarm
	Delete all recovered alarm
	Auto run ON, OFF
	Display Auto run
	Alarm occurred
	Move to right page
	Move to left page
	Move to upper page
	Move to below page

Handling method

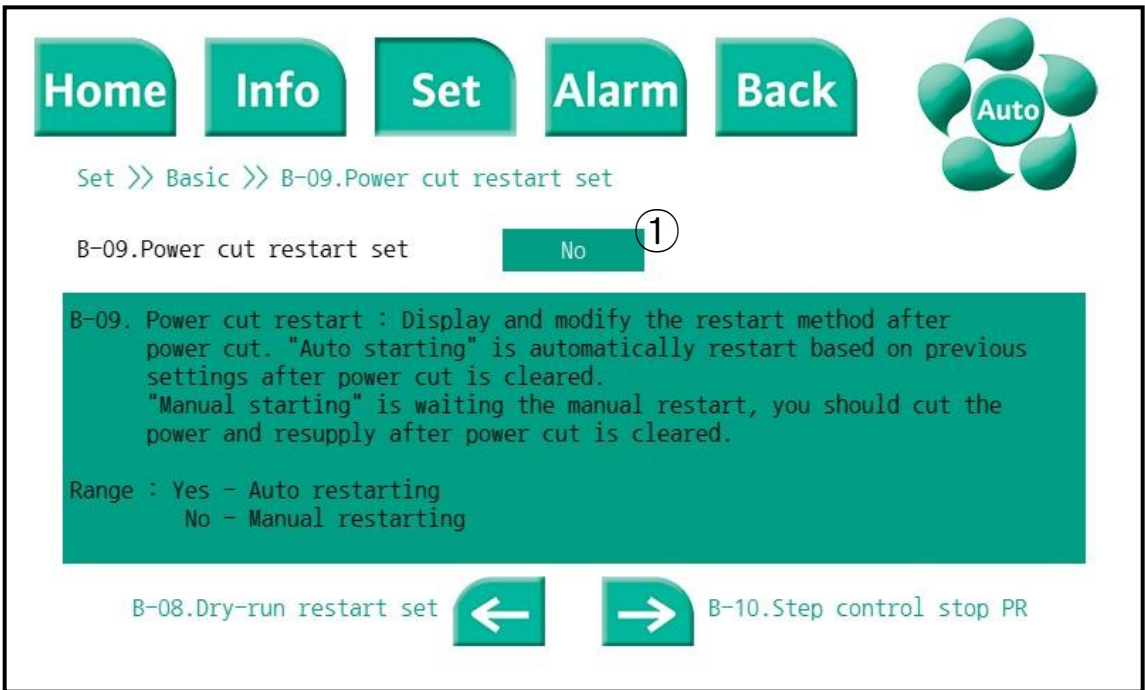
1. How to use pop-up screen



- 1) If you want to change "B-01. Set PR", touch "①". Pop-up screen "②" turn up.
- 2) Enter the value by "②" keypad.
- 3) On the keypad 'Esc' means cancel, '←' means delete 1 digit, 'Clear' means delete all, '↵' means save.
- 4) If save the value or cancel the value, pup-up screen will turn off.
- 5) Can't input the outside of setting range. When you enter the value, refer to help.

Handling method

2. How to use touch screen



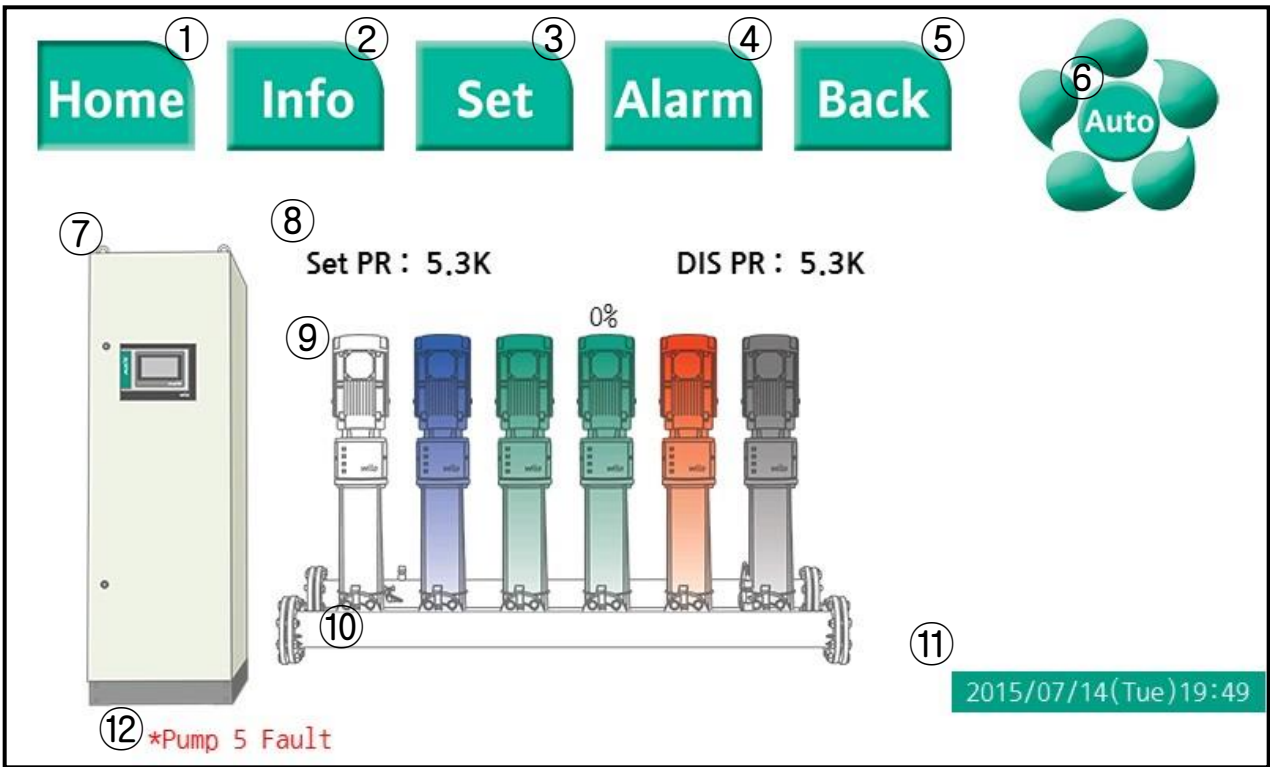
- 1) If you want to change "B-09. Power cut restart", touch "①".
The value change "Yes" or "No".
- 2) Back to original value, touch "①" 1 more.
- 3) The value will save automatically without save action.

Menu tree

Home	Home	Page. 31
	Installation help	Page. 33
Info	System	Page. 34
	Pressure Trigger	Page. 35
	Pump 1	Page. 35
	Pump 2	
	Pump 3	
	⋮	
	Pump 6	
Set	Basic	
	B-01. Set PR	Page. 37
	B-02. Shut PR	Page. 37
	B-03. Starting PR	Page. 38
	B-04. Final stop PR	Page. 38
	B-05. MIN run time	Page. 38
	B-06. FC MIN FREQ	Page. 39
	B-07. Sequence delay time	Page. 39
	B-08. Dry-run restart set	Page. 39
	B-09. Power cut restart set	Page. 40
	B-10. Step control stop PR	Page. 40
	Extra	
	E-01. MAX PR	Page. 42
	E-02. MIN PR	Page. 42
	E-03. MIN PR DTC delay	Page. 42
	E-04. MIN PR set	Page. 43
	E-05. MIN SUC PR limit	Page. 43
	E-06. SUC PR DTC delay	Page. 43
	E-07. Sensor Offset	Page. 44
	E-08. Sensor Span	Page. 44
	E-09. Shut PR deviation	Page. 45
	E-10. Shut PR DTC time	Page. 45
	E-11. Shut PR judging time	Page. 45
	E-12. Sampling cycle	Page. 46
	E-13. PID Value	Page. 46
	E-14. Starting number	Page. 46

Menu tree

Extra	E-15. Dry-run DTC delay	Page. 47	
	E-16. Loss compensate	Page. 47	
	E-17. Reduction set	Page. 47	
	E-18. Pump block DTC time	Page. 48	
	E-19. Pump block run time	Page. 48	
	E-20. Step changeover	Page. 48	
	E-21. Pipefill PR	Page. 49	
	E-22. Pipefill end DTC time	Page. 49	
	E-23. Low-water LV DTC	Page. 49	
	E-24. Low-water LV DTC delay	Page. 50	
	E-25. Low-water LV RSTR set	Page. 50	
	E-26. Backlight time	Page. 50	
	Factory	F-01. Sub DIS PT	
		F-02. SUC PT	
		F-03. Register Sub pump	
		F-04. Remote control ADD	
		F-05. Cut-in FREQ	
		F-06. Unit change	
		F-07. FC change cycle	
		F-08. Auto operating mode	
		F-09. Date	
		F-10. Pump initialization	
		F-11. System initialization	
		F-12. Password	
		F-13. DIS PT capacity	
		F-14. Half FREQ conversion rate	
Schedule	S-01. Monthly Schedule	Page. 52	
	S-02. Daily Schedule	Page. 53	
	S-03. Hourly Schedule	Page. 53	
	S-04. Seasonal Schedule	Page. 54	
Alarm	Alarm list and Initialization	Page. 55	



PUZeN e home screen.

- ① Home Icon : Move to "Home" screen
- ② Info Icon : Move to "Info" screen
- ③ Set Icon : Move to "Set" screen
- ④ Alarm Icon : Move to "Alarm" screen
- ⑤ Back Icon : Move to previous menu

- ⑥  :  Touch → Auto run start(Water drop animation rotate)
-  :  Retouch → Auto run stop(Water drop animation rotation stop)
-  : Alarm occurred  →  (Water drop animation rotate)
-  : Touch  → Auto run Stop(Water drop animation rotation stop)

- ⑦ Panel Graphic : Move to "Info>>System" screen directly(refer to page 33)
- ⑧ Set PR & DIS PR : Move to "Set>>Basic>B-01. Set PR" screen directly by touching "Set PR"
Move to "Info>>Pressure Trigger" screen directly by touching "DIS PR"

Home

Home

- ⑨  : Green – Pump run by common power
Red – Pump fault
White – Pump stand by
Blue – Pump set by manual mode
Gray – Pump set by reserved mode(Back up)

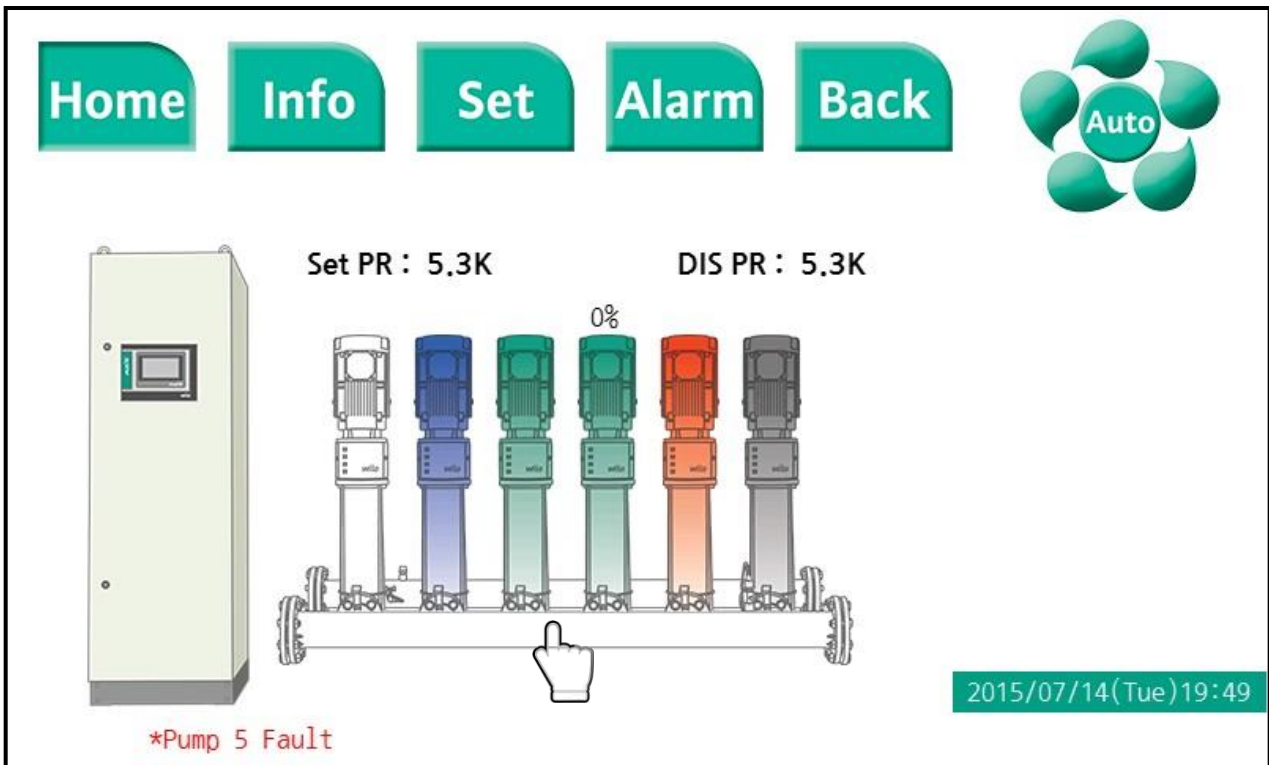
100%



- : Green and pump load rate display, Indicates pump run by frequency converter (40% ~ 100%, Display by 1% unit)
Move to "Info>>Pump #" screen directly(refer to page 34)

- ⑩ Manifold Graphic : Move to "Installation help" directly(refer to page 31)
⑪ Date, day, time display : Move to "Set>>Factory>>F-11.Date" screen directly
⑫ Alarm message

Installation help



Move to installation help page, when touch the manifold at "Home" display.

Home

Info

Set

Alarm

Back



Home >> Installation Help

1. Check the state of main power supply.

- Check the state of circuit braker and protector capacity.
- Check the voltage of main power supply

2. Check the level of water tank.

- Recommended water level of tank is over than 2m

Prev





Next

Home

Info

Set

Alarm

Back



Home >> Installation Help

3. Check the condition of suction side. If suction side's condition is not suitable, booster system can not operate normally

- In case the suction pipe's length is over than 10m or pipe is bent more than 2 times, suction pipe should be one size bigger than booster system manifold

4. Expel air from the pumps and pipes

- Expel air from the pipes of installation site, pipe of booster system and pumps

Prev





Next

Home

Info

Set

Alarm

Back



Home >> Installation Help

5. Check the rotation of pumps.

- Check the rotation of pumps one by one.(The rotating direction of a pump is marked on a fan cover of motor)
- If the rotation of pump is reversed, change the 2 phase of the motor.

6. Setup operating data of booster system appropriately.

- Setup operating data based on name plate of booster system and installation site's condition.
- Read the [Set display] part of KF Touch operation instruction before setting up the operating data.

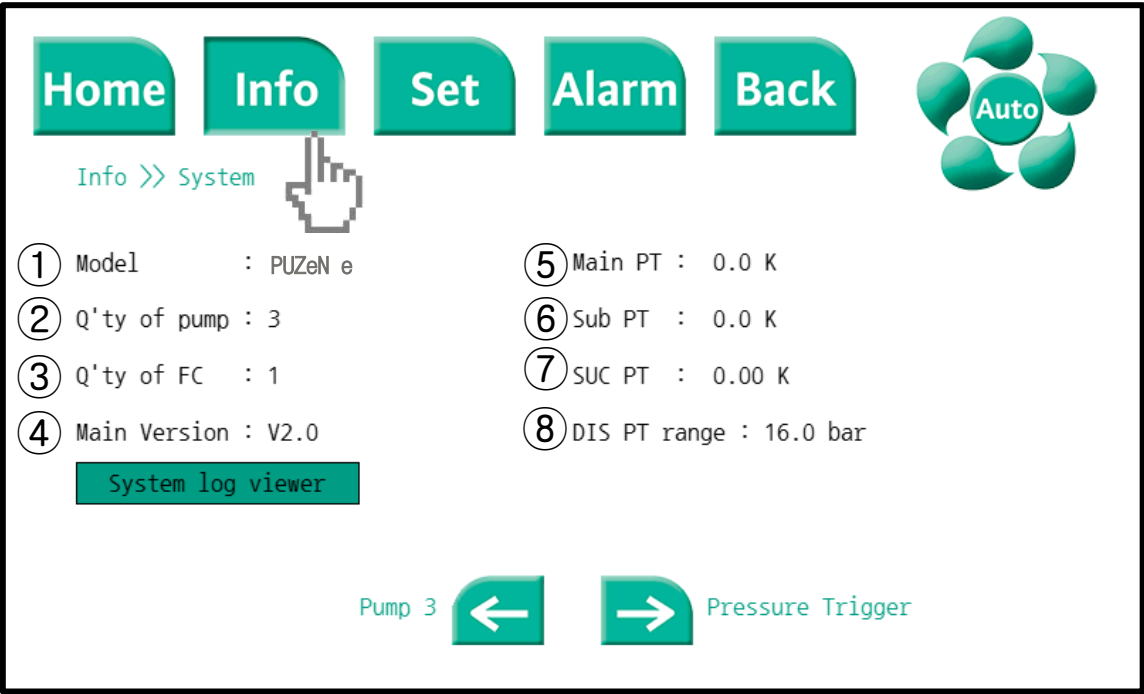
7. Move to the "Home" display when you touch "End" icon

Prev





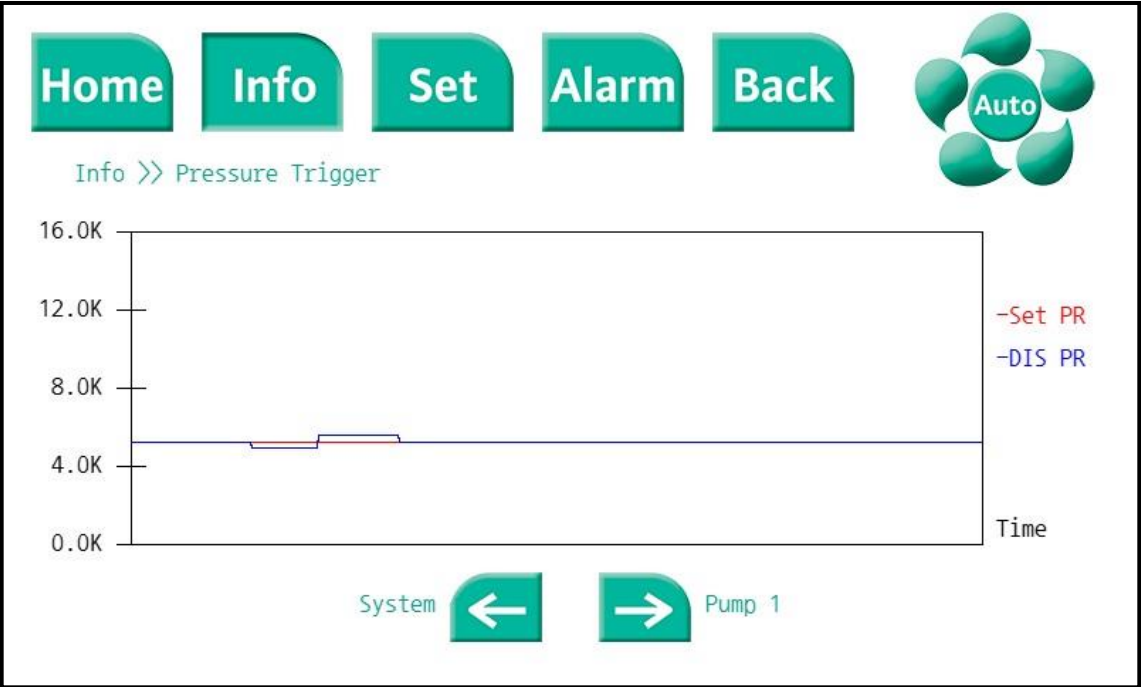
Done



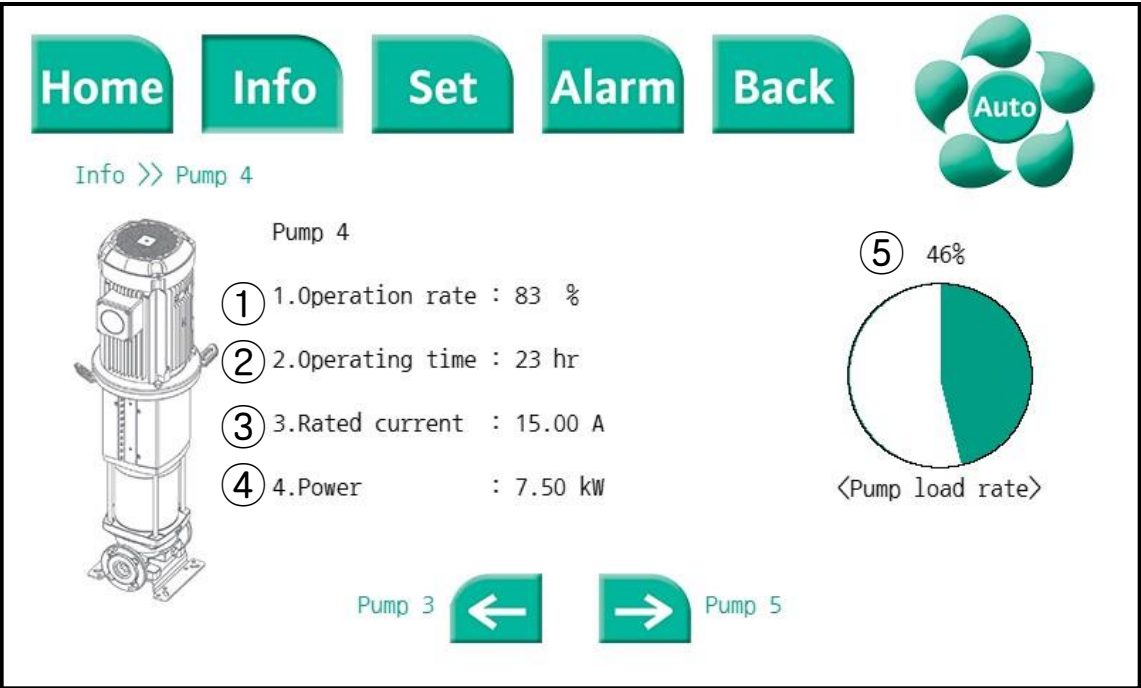
This is the system information screen("Info" icon touch), display the information of PUZeN e booster system

- ① Booster system model name
- ② Quantity of booster system's pumps
- ③ Quantity of booster system's frequency converter
- ④ TCU G22 program version
- ⑤ Display Main Pressure transmitter(Discharge)
- ⑥ Display Sub(spare) pressure transmitter(Discharge)
- ⑦ Display Suction pressure transmitter
- ⑧ Display Pressure transmitter range

Info >> Pressure trigger



Pressure trigger : Show the set pressure and discharge pressure by graph.
Time cycle is 60 sec.



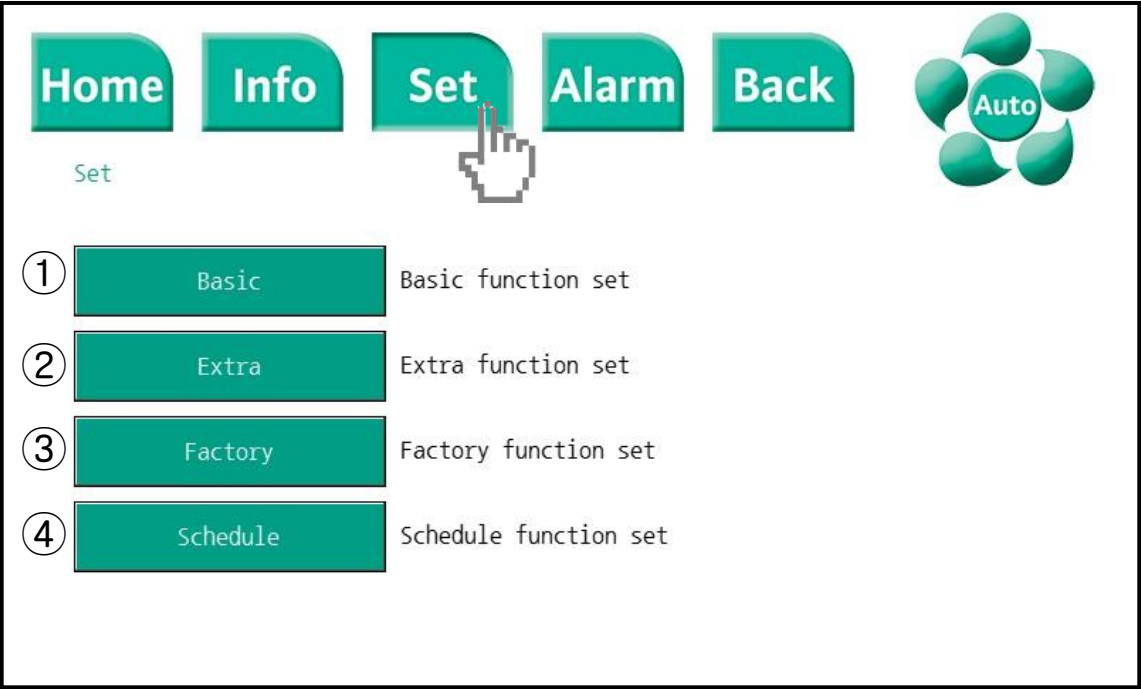
Display information of Pump 4

- ① Display Pump 4 operation rate
- ② Display Pump 4 operating time
- ③ Display Pump 4 rated current
- ④ Display Pump 4 power
- ⑤ Display Pump 4 load rate

※ This is example of 6 pumps booster system.

Set

Set



Set screen(Touch "Set" icon)

- ① Move to Basic function set
- ② Move to Extra function set
- ③ Move to Factory function set
- ④ Move to Schedule function set

Set

Basic

Home

Info

Set

Alarm

Back



Set >> Basic

B-01.Set PR

B-02.Shut PR

B-03.Starting PR

B-04.Final stop PR

B-05.MIN run time

B-06.FC MIN FREQ

B-07.Sequence delay time

B-08.Dry-run restart set

B-09.Power cut restart set

B-10.Step control stop PR

Home

Info

Set

Alarm

Back



Set >> Basic >> B-01.Set PR

B-01.Set PR

5.3

K

B-01. Set PR : Display and modify demanding pressure on the site where booster system is installed.
Range : 0.6 ≤ Set PR ≤ 9.9 K

B-10.Step control stop PR

←

→

B-02.Shut PR

Home

Info

Set

Alarm

Back



Set >> Basic >> B-02.Shut PR

B-02.Shut PR

10.6

K

B-02. Shut PR : Display and modify shut off pressure of pump.
Range : Set PR+0.7 ≤ Shut PR ≤ MAX PT

B-01.Set PR

←

→

B-03.Starting PR

Set

Basic

Home

Info

Set

Alarm

Back



Set >> Basic >> B-03.Starting PR

B-03.Starting PR

5.1

K

B-03. Starting PR : Display and modify pressure which starts to operate Initial pump during frequency converter control mode.
(This function doesn't support on step control mode)

Range : Set PR-0.5 ≤ Starting PR ≤ Set PR
Set PR-0.5 ≥ 0.1

B-02.Shut PR

←

→

B-04.Final stop PR

Home

Info

Set

Alarm

Back



Set >> Basic >> B-04.Final stop PR

B-04.Final stop PR

5.6

K

B-04. Final stop PR : Display and modify final stop pressure for booster system.

Range : Set PR+0.2 ≤ Fianl stop PR ≤ Set PR+0.5
Fianl stop PR ≤ Shut PR-0.2

B-03.Starting PR

←

→

B-05.MIN run time

Home

Info

Set

Alarm

Back



Set >> Basic >> B-05.MIN run time

B-05.MIN run time

5

sec

B-05. MIN run time : Display and modify minimum run time of pump pump operates.

Range : 1sec ≤ MIN run time ≤ 30sec

B-04.Final stop PR

←

→

B-06.FC MIN FREQ

Set

Basic

HomeInfoSetAlarmBack

Auto

Set >> Basic >> B-06.FC MIN FREQ

B-06.FC MIN FREQ20Hz

B-06. FC MIN FREQ : Display and modify minimum frequency of frequency converter during the frequency converter control mode.
(This function doesn't support on step control mode)

Range : 20Hz ≤ FC MIN FREQ ≤ 55Hz

B-05.MIN run time←→B-07.Sequence delay time

HomeInfoSetAlarmBack

Auto

Set >> Basic >> B-07.Sequence delay time

B-07.Sequence delay time5sec

B-07. Sequence delay time : Display and modify delay time of pump cut-in and cut-out.

Range : 1sec ≤ Sequence delay time ≤ 10sec

B-06.FC MIN FREQ←→B-08.Dry-run restart set

HomeInfoSetAlarmBack

Auto

Set >> Basic >> B-08.Dry-run restart set

B-08.Dry-run restart setNo

B-08. Dry-run restart set : Display and modify the restart method after dry-run is cleared. "Auto starting" is automatically restart based on previous settings after dry run is cleared. "Manual starting" is waiting the manual restart, you should cut the power and resupply after dry run is cleared.

Range : Yes - Auto restarting No - Manual restarting

*Booster system stop regardless of "Auto starting" or "Manual starting" when dry run continually occurs 5 times.

B-07.Sequence delay time←→B-09.Power cut restart set

Set

Basic

Home

Info

Set

Alarm

Back



Set >> Basic >> B-09.Power cut restart set

B-09.Power cut restart set

No

B-09. Power cut restart : Display and modify the restart method after power cut. "Auto starting" is automatically restart based on previous settings after power cut is cleared.
"Manual starting" is waiting the manual restart, you should cut the power and resupply after power cut is cleared.

Range : Yes - Auto restarting
No - Manual restarting

B-08.Dry-run restart set





B-10.Step control stop PR

Home

Info

Set

Alarm

Back



Set >> Basic >> B-10.Step control stop PR

B-10.Step control stop PR

5.8 K

B-10. Step control stop PR : Display and modify stop pressure in case of step control mode due to frequency converter fault.

Range : Set PR+0.5 ≤ Step control stop PR ≤ Set PR+3.0
Step control stop PR ≤ Shut PR-0.2

B-09.Power cut restart set





B-01.Set PR

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra

E-01.MAX PR

E-02.MIN PR

E-03.MIN PR DTC delay

E-04.MIN PR set

E-05.MIN SUC PR limit

E-06.SUC PR DTC delay

E-07.Sensor Offset

E-08.Sensor Span

E-09.Shut PR deviation

E-10.Shut PR DTC time

←

→

E-11 ~ E-20

HomeInfoSetAlarmBack

Auto

Set >> Extra

E-11.Shut PR judging time

E-12.Sampling cycle

E-13.PID Value

E-14.Starting number

E-15.Dry-run DTC delay

E-16.Loss compensate

E-17.Reduction set

E-18.Pump block DTC time

E-19.Pump block run time

E-20.Step changeover

E-01 ~ E-10

←

→

E-21 ~ E-26

HomeInfoSetAlarmBack

Auto

Set >> Extra

E-21.Pipefill PR

E-22.Pipefill end DTC time

E-23.Low-water LV DTC

E-24.Low-water LV DTC delay

E-25.Low-water LV RSTR set

E-26.Backlight time

E-11 ~ E-20

←

→

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-01.MAX PR

E-01.MAX PR16.0K

E-01. MAX PR : Display and modify value of abnormal high discharge pressure.
Range : Shut PR < MAX PR ≤ MAX DIS PT

E-10.Shut PR DTC time←→E-02.MIN PR

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-02.MIN PR

E-02.MIN PR20%1.1K

E-02. MIN PR : Display and modify value of abnormal low discharge pressure. MIN PR alarm occur when discharge pressure is lower than "E-02. MIN PR".
Range : 0 ≤ MIN PR ≤ 90% of set PR

E-01.MAX PR←→E-03.MIN PR DTC delay

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-03.MIN PR DTC delay

E-03.MIN PR DTC delay5sec

E-03. MIN PR DTC delay : Display and modify delay time for detecting "E-02. MIN PR".
Range : 1sec ≤ MIN PR DTC delay ≤ 30sec

E-02.MIN PR←→E-04.MIN PR set

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-04.MIN PR set

E-04.MIN PR set

No

E-04. MIN PR set : Choose to stop or keep running booster system when abnormal low discharge pressure occurs.
Range : Yes - Booster system stops running when abnormal low discharge pressure occurs.
no - Booster system makes an alarm and keeps running when abnormal low discharge pressure occurs.

E-03.MIN PR DTC delay

←

→

E-05.MIN SUC PR limit

HomeInfoSetAlarmBack

Auto

Set >> Extra >> B-05.MIN run time

E-05.MIN SUC PR limit

0.10

K

E-05. MIN SUC PR limit : Display and modify value of minimum pressure on suction side of booster system. If pressure on suction side of booster system is lower than "E-05. MIN SUC PR limit", alarm will occur
Range : SUC PT 16bar - 0 ≤ MIN SUC PR limit ≤ 9.99bar
SUC PT 1bar - 0 ≤ NMIN SUC PR limit ≤ 1.00bar

E-04.MIN PR set

←

→

E-06.SUC PR DTC delay

HomeInfoSetAlarmBack

Auto

Set >> Extra >> B-06.FC MIN FREQ

E-06.SUC PR DTC delay

5

sec

E-06. SUC PR DTC delay : Display and modify delay time for detecting "E-05. MIN SUC PR limit".
Range : 1sec ≤ SUC PR DTC delay ≤ 30sec

E-05.MIN SUC PR limit

←

→

E-07.Sensor Offset

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-07.Sensor Offset

E-07.Sensor Offset

0.0

K

E-07. Sensor offset : Display and modify the discharge pressure transmitter offset

Range : -0.5K ≤ Sensor offset ≤ +0.5K

E-06.SUC PR DTC delay

←

→

E-08.Sensor Span

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-08.Sensor Span

E-08.Sensor Span

100

%

E-08. Sensor span : Display and modify the discharge pressure transmitter span

Range : 95% ≤ Sensor span ≤ 105%

E-07.Sensor Offset

←

→

E-09.Shut PR deviation

Set

Extra

Home

Info

Set

Alarm

Back



Set >> Extra >> E-09.Shut PR deviation

E-09.Shut PR deviation

0.2

K

E-09. Shut PR deviation : Display and modify shut off pressure deviation for booster system stop in frequency converter control mode.

Range : 0.0K ≤ Shut PR deviation ≤ 0.5K

E-08.Sensor Span

←

→

E-10.Shut PR DTC time

Home

Info

Set

Alarm

Back



Set >> Extra >> E-10.Shut PR DTC time

E-10.Shut PR DTC time

40

sec

E-10. Shut PT DTC time : Display and modify shut off pressure detecting time for booster system stop in frequency converter control mode.

Range : 20sec ≤ Shut PR DTC time ≤ 60sec

E-09.Shut PR deviation

←

→

E-11.Shut PR judging time

Home

Info

Set

Alarm

Back



Set >> Extra >> E-11.Shut PR judging time

E-11.Shut PR judging time

5

sec

E-11. Shut PR judging time : Display and modify shut off pressure judging time for booster system stop in frequency converter control mode.

Range : 2sec ≤ Shut PR judging time ≤ 15sec

E-10.Shut PR DTC time

←

→

E-12.Sampling cycle

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-12.Sampling cycle

E-12.Sampling cycle10ms

E-12. Sampling cycle : Display and modify the sampling cycle.
Range : 10ms ≤ Sampling cycle ≤ 2000ms(displacement 10ms)
※1s = 1000ms

E-11.Shut PR judging time←→E-13.PID Value

- This value can effect performance of control panel
- Before make a change, check the installation site environments.

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-13.PID Value

E-13.PID ValueP5I110D0

E-13. PID Value : Display and modify the P, I, D value
Range : 10 ≤ P,I,D Value ≤ 100

E-12.Sampling cycle←→E-14.Starting number

- This value can effect performance of control panel
- Before make a change, check the installation site environments.

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-14.Starting number

E-14.Starting number10times

E-14. Starting number : Display and modify pump starting number.
Make an alarm when number of pump starting is more than
"E-14. Starting number" in an hour.
Range : 0time ≤ Starting number ≤ 100times
※Set 0 means disuse this function.

E-13.PID Value←→E-15.Dry-run DTC delay

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-15.Dry-run DTC delay

E-15.Dry-run DTC delay5sec

E-15. Dry-run DTC delay : Display and modify delay time for detecting dry-run
Range : 1sec ≤ Dry-run DTC delay ≤ 30sec

E-14.Starting number←→E-16.Loss compensate

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-16.Loss compensate

E-16.Loss compensate100%5.3K

E-16. Loss compensate : Display and modify compensation rate for friction loss.
Range : 30% ≤ Loss compensate ≤ 100%

E-15.Dry-run DTC delay←→E-17.Reduction set

- This value can effect control panel’s performance.
- Before the change default, check the installation site environments.

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-17.Reduction set

E-17.Reduction set0

E-17. Reduction set : Display and modify number of pump to exclude from operating pumps in booster system. Excluded pumps don't run even other pumps fault.(This value can't change on auto operating mode)
Range : 0 ≤ Reduction set ≤ Booster system pump number - 1

E-16.Loss compensate←→E-18.Pump block DTC time

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-18.Pump block DTC time

E-18.Pump block DTC time24Hr

E-18. Pump block DTC time : Display and modify delay time for detecting pump block in auto operating mode.
Range : 0 / 12hr ≤ Pump block DTC time ≤ 72hr

E-17.Reduction set ← → E-19.Pump block run time

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-19.Pump block run time

E-19.Pump block run time2sec

E-19. Pump block run time : Display and modify run time to prevent pump block. If pump has no operating time during "E-18.Pump block DTC time" the pumps will operate for setting time of "E-19.Pump block run time".
Range : 1sec ≤ Pump block run time ≤ 30sec

E-18.Pump block DTC time ← → E-20.Step changeover

- This value can effect control panel's performance.
- Before the change default, check the installation site environments.

HomeInfoSetAlarmBack

Auto

Set >> Extra >> E-20.Step changeover

E-20.Step changeoverNo

E-20. Step changeover : If frequency converter is out of order in Uni frequency converter operating mode, operating mode will change automatically to step control or not.
(This value can't change on auto operating-mode)
Range : Yes - Changeover to step control mode.
No - Do not changeover to step control mode.

E-19.Pump block run time ← → E-21.Pipefill PR

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra >>

E-21.Pipefill PR60%3.2K

E-21. Pipefill PR : Display and modify value for determining pipefill complete. Pipefill is completed when DIS PR exceed pipefill PR.
Range : 0 ≤ Pipefill PR ≤ 90%
※0% - disuse pipefill

E-20.Step changeover←→E-22.Pipefill end DTC time

HomeInfoSetAlarmBack

Auto

Set >> Extra >>

E-22.Pipefill end DTC time10sec

E-22. Pipefill end DTC time : Display and modify delay time for returning to normal operation after pipefill accelation complete.
Range : 1sec ≤ Pipefill end DTC time ≤ 60sec

E-21.Pipefill PR←→E-23.Low-water LV DTC

- This value can effect control panel’s performance.
- Before the change default, check the installation site environments.

HomeInfoSetAlarmBack

Auto

Set >> Extra >>

E-23.Low-water LV DTCNo

E-23. Low-water LV DTC : Display and modify use or disuse low-water level detection.
Range : Yes / No

E-22.Pipefill end DTC time←→E-24.Low-water LV DTC delay

Set

Extra

HomeInfoSetAlarmBack

Auto

Set >> Extra >>

E-24.Low-water LV DTC delay10sec

E-24. Low-water LV DTC delay : Display and modify delay time for detecting low-water level.
Range : 1 ≤ Low-water LV DTC time ≤ 30 sec

E-23.Low-water LV DTC←→E-25.Low-water LV RSTR set

HomeInfoSetAlarmBack

Auto

Set >> Extra >>

E-25.Low-water LV RSTR setNo

E-25. Low-water LV RSTR set : Display and the restart method after low-water level detection. "Auto starting" is automatically restart based on previous settings after low-level is cleared.
"Manual starting" is waiting the manual restart.
User should cut the power and resupply after low-level is cleared.
Range : Yes - Auto restarting No - Manual restarting
※Booster system stop regardless of "Auto starting" or "Manual starting" when low-water level continually occurs 5 times.

E-24.Low-water LV DTC delay←→E-26.Backlight time

HomeInfoSetAlarmBack

Auto

Set >> Extra >>

E-26.Backlight time5min

E-26. Backlight time : Display and modify delay time for turning off backlight
Backlight will not turn off when system is having any fault.
Range : 0 ≤ Backlight time ≤ 30 minutes
※0 - backlight always on

E-25.Low-water LV RSTR set←→E-01.MAX PR

Set

Factory

HomeInfoSetAlarmBack

Auto

<Password>

56789

01234

<-ClrEnt

- Password is needed when entering the factory menu.
- Factory menu is already set by WILO, changing factory menu can cause wrong behavior and fault. Before changing factory menu, ask to WILO service.

HomeInfoSetAlarmBack

Auto

Set >> Factory

F-01.Sub DIS PT

F-02.SUC PT

F-03.Reserved pump

F-04.Remote control ADD

F-05.Cut-in FREQ

F-06.Unit change

F-07.FC change cycle

F-08.Auto operating mode

F-09.Date

F-10.Pump initialization

<

>

F-11 ~ F-14

HomeInfoSetAlarmBack

Auto

Set >> Factory

F-11.Date

F-12.Pump initialization

F-13.System initialization

F-14.Password

F-15.DIS PT range

F-16.Half FREQ ratio

F-17.Half power

F-18.Half rated current

F-01 ~ F-10

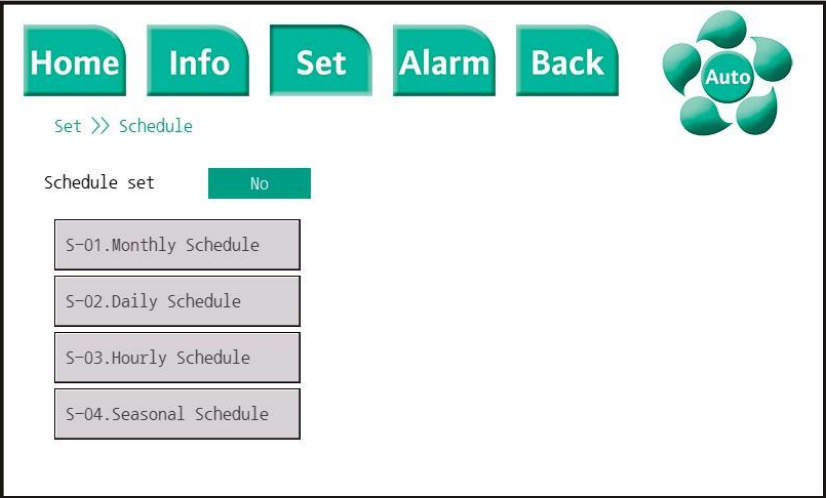
<

>

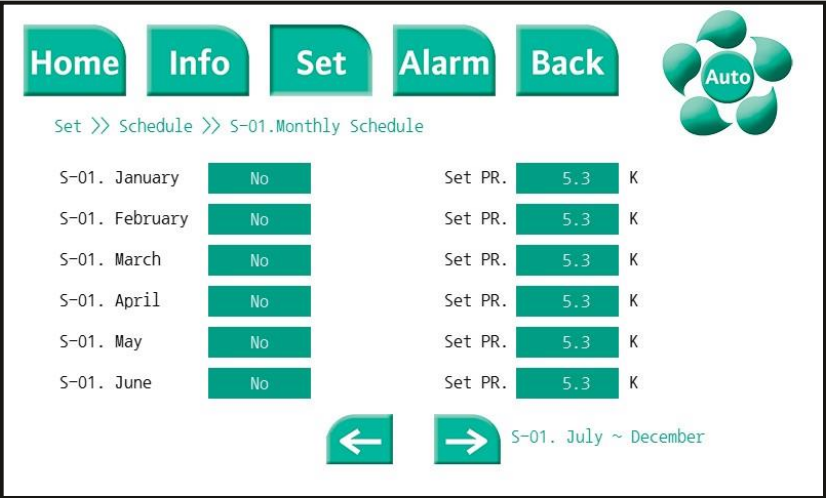
- F-16, F-17, F-18 are enabled in PZEH

Set

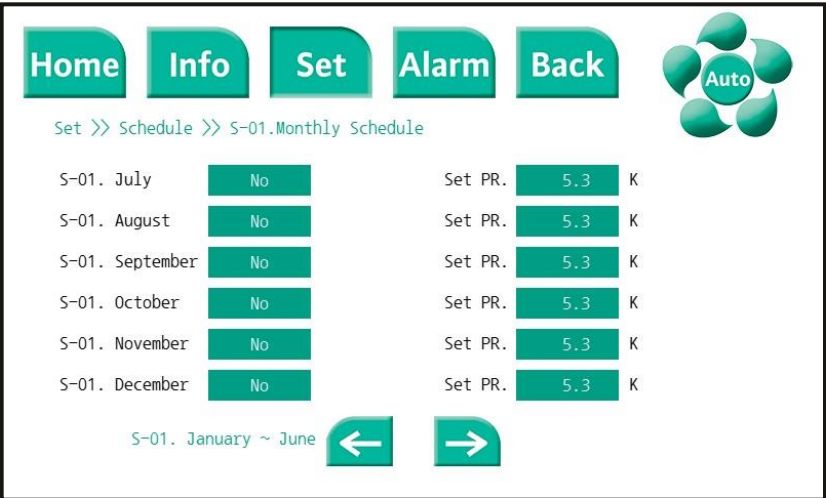
Schedule



- Touch "①", menu will change "No"/"Monthly"/"Daily"/"Hourly"/"Seasonal".
- Activate menu's color will change and touch the menu for detail schedule set.



- Touch the menu to set the schedule, after "No" → "Yes" set the setting pressure.



Set

Schedule

HomeInfoSetAlarmBack

Auto

Set >> Schedule >> S-02.Daily Schedule

S-02. Monday	No	Set PR.	5.3	K
S-02. Tuesday	No	Set PR.	5.3	K
S-02. Wednesday	No	Set PR.	5.3	K
S-02. Thursday	No	Set PR.	5.3	K
S-02. Friday	No	Set PR.	5.3	K
S-02. Saturday	No	Set PR.	5.3	K
S-02. Sunday	No	Set PR.	5.3	K

HomeInfoSetAlarmBack

Auto

Set >> Schedule >> S-03.Hourly Schedule

S-03. 0 o'clock	No	Set PR.	5.3	K
S-03. 1 o'clock	No	Set PR.	5.3	K
S-03. 2 o'clock	No	Set PR.	5.3	K
S-03. 3 o'clock	No	Set PR.	5.3	K
S-03. 4 o'clock	No	Set PR.	5.3	K
S-03. 5 o'clock	No	Set PR.	5.3	K

S-03. 18 o'clock ~ 23 o'clock

←

→

S-03. 6 o'clock ~ 11 o'clock

HomeInfoSetAlarmBack

Auto

Set >> Schedule >> S-03.Hourly Schedule

S-03. 6 o'clock	No	Set PR.	5.3	K
S-03. 7 o'clock	No	Set PR.	5.3	K
S-03. 8 o'clock	No	Set PR.	5.3	K
S-03. 9 o'clock	No	Set PR.	5.3	K
S-03. 10 o'clock	No	Set PR.	5.3	K
S-03. 11 o'clock	No	Set PR.	5.3	K

S-03. 0 o'clock ~ 5 o'clock

←

→

S-03. 12 o'clock ~ 17 o'clock

Set

Schedule

HomeInfoSetAlarmBack

Auto

Set >> Schedule >> S-03.Hourly Schedule

S-03. 12 o'clock	No	Set PR.	5.3	K
S-03. 13 o'clock	No	Set PR.	5.3	K
S-03. 14 o'clock	No	Set PR.	5.3	K
S-03. 15 o'clock	No	Set PR.	5.3	K
S-03. 16 o'clock	No	Set PR.	5.3	K
S-03. 17 o'clock	No	Set PR.	5.3	K

S-03. 6 o'clock ~ 11 o'clock

←

→

S-03. 18 o'clock ~ 23 o'clock

HomeInfoSetAlarmBack

Auto

Set >> Schedule >> S-03.Hourly Schedule

S-03. 18 o'clock	No	Set PR.	5.3	K
S-03. 19 o'clock	No	Set PR.	5.3	K
S-03. 20 o'clock	No	Set PR.	5.3	K
S-03. 21 o'clock	No	Set PR.	5.3	K
S-03. 22 o'clock	No	Set PR.	5.3	K
S-03. 23 o'clock	No	Set PR.	5.3	K

S-03. 12 o'clock ~ 17 o'clock

←

→

S-03. 0 o'clock ~ 5 o'clock

HomeInfoSetAlarmBack

Auto

Set >> Schedule >> S-04.Seasonal Schedule

S-04. Spring	No	Set PR.	5.3	K
S-04. Summer	No	Set PR.	5.3	K
S-04. Autumn	No	Set PR.	5.3	K
S-04. Winter	No	Set PR.	5.3	K

Spring : March / April / May

Summer : June / July / August

Fall : September / October / November

Winter : December / January / February

Alarm

Alarm

HomeInfoSetAlarmBack

Alarm

①

②

③

Time	Alarm message	Recovery
2015/07/14(Tue)19:58	Pump 5 Fault	
2015/07/14(Tue)19:58	Dry-running	2015/07/14(Tue)19:58
2015/07/14(Tue)19:58	Low-level	2015/07/14(Tue)19:58
2015/07/14(Tue)19:58	Max. pressure	2015/07/14(Tue)19:58
2015/07/14(Tue)19:58	FC1 Fault	
2015/07/14(Tue)19:58	FC2 Fault	

Auto

④

⑤

⑥

⑦

Del

Reset

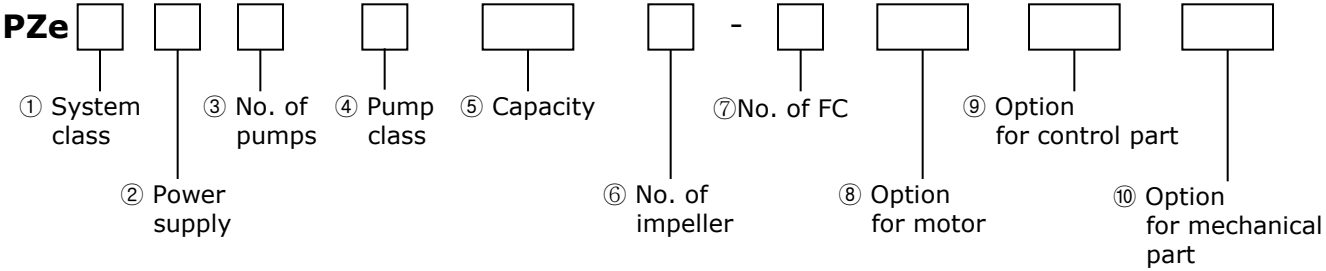
*FC2 Fault *FC1 Fault *Pump 5 Fault

Maximum 37 alarm message can save.

- ① Show the occurred time of alarm
- ② Show the message of alarm
- ③ Show the recovery time of alarm, if the alarm is not recovered display continuous
- ④ Move to upper page, when
- ⑤ Move to downer
- ⑥ Delete each recovered alarm
- ⑦ Delete all recovered alarm

- ※ Alarm message list
- Pump # Fault
 - Dry-running
 - Repeated dry
 - FC# Fault
 - Low discharge pressure
 - Setting values
 - PT Fault
 - Max. Pressure
 - Repeated Max. pressure
 - Sub PT Fault
 - Main PT Fault
 - 10% PTs Error
 - Low suction pressure
 - # starting No.

Designation of PUZeN e booster system



① System class

Code	System class
PZEU	PUZeN e Uni(Single Inverter)
PZEC	PUZeN e Combi System
PZEH	PUZeN e Half System
PZEM	PUZeN e Multi(Multiple Inverter)

② Power supply

Code	Power	Code	Power
2	220V,60Hz	B	220V,50Hz
3	380V,60Hz	C	380V,50Hz
4	440V,60Hz	D	440V,50Hz
5	460V,60Hz	E	460V,60Hz
6	480V,60Hz	F	480V,50Hz

③ No. of pumps

Code	No. of pumps
2	2 pcs
3	3 pcs
4	4 pcs
5	5 pcs
6	6 pcs

④ Pump class

Code	Pump class
H	Helix V Series
M	MVI Series

⑤ Capacity
- Pump series

⑥ No. of impeller

Code	No. of impeller	Code	No. of impeller	Code	No. of impeller
01	1 stg	06	6 stg	10	11 stg
02	2 stg	07	7 stg	11	12 stg
03	3 stg	08	8 stg	12	13 stg
04	4 stg	09	9 stg	13	14 stg
05	5 stg	A	10 stg	14	15 stg

⑦ No. of FC

Code	No. of pumps
1	1 pcs
2	2 pcs
3	3 pcs
4	4 pcs
5	5 pcs
6	6 pcs

⑧ Option for motor

Code	Option
E2	IE2 Motor
E3	IE3 Motor
G3	Explosion proof Motor

⑨ Option for control part

Code	Option
S	Stainless steel Panel
L	Individual Earth Leakage Breaker
E	EOCR(Electronic Over Current Relay)

⑩ Option for mechanical part

Code	Option	Code	Option
1	Bigger bore size for suction pipe	K	Forging Counter Flange
2	Bigger bore size for discharge pipe	O	Remove 1 impeller stage
3	STS304 bed	T	Applying SS bed
I	Insulation bolt and nut		